Lifestyle Factors and Well-Being

An Exploration of Factors Impacting Well-Being

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Abstract

This study examined the effect of lifestyle factors such as sleep, exercise, nutrition, social media use, social connectedness, stress management, and time spent in nature on well-being. 51 participants completed a survey through Prolific, where they self-reported their habits in each of these categories of lifestyle factors. They also rated their subject well-being and life satisfaction. A multiple linear regression was conducted exploring, establishing significant positive effects of sleep, belonging, and stress management skills on self-reported well-being $F(9, 41) = 10.7$, $p < .001$. Participants who reported better sleep, stronger feelings of social connectedness, and greater ability to manage stress also reported higher levels of well-being. These findings highlight the importance of promoting healthy sleep patterns, fostering social connections, and building effective stress management strategies to enhance psychological well-being. Implications for wellness interventions and public health initiatives are discussed.

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1 Introduction

There exists a substantial collection of research regarding the impact of lifestyle factors on well-being and life satisfaction. Researchers have explored lifestyle factors such as sleep, diet, exercise, loneliness, and perceived pressure, and their effect on self-reported well-being. One study found that sleep has a significant main effect on stress and well-being, such that the effect on stress on subjective well-being can be reduced when adequate sleep quality is acheived (Weinberg et al., 2016). Another study found that consumption of a well-balanced and varied diet, accompanied with a healthy lifestyle, plays a critical role in preservation of normal brain function and mental well-being (Muscaritoli, 2021). Furthermore, a study conducted in China found that loneliness and perceived pressure had statistically significant negative associations with mental well-being (Wang et al., 2023). In regards to social media use, one study found that social media can either enhance or diminish well-being, depending on how it is used by the consumer (Kross et al., 2021). Finally, one study found that increasing nature connectedness was positively correlated with both psychological and social well-being (Howell et al., 2011).

Considering the extensive research that has been conducted thus far on lifestyle factors and their impact on well-being, this study seeks to consolidate this research into one comprehensive study through a survey study design. It is hypothesized that factors such as sleep, exercise, feeling a sense of belonging, time spent in nature, and stress management skills will be positively correlated with well-being and that factors such as social media use and feeling constantly under pressure will be negatively correlated with participant well-being.

```
# notebook setup
options(conflicts.policy = "depends.ok")
library(tidyverse)
library(skimr)
library(readr)
library(cowplot, include.only = c("plot_grid", "theme_half_open"))
library(corrplot, include.only = "corrplot.mixed")
library(broom)
library(ggplot2)

devtools::source_url("https://github.com/jjcurtin/lab_support/blob/main/fun_eda.R?raw=true")
devtools::source_url("https://github.com/jjcurtin/lab_support/blob/main/fun_plots.R?raw=true")
devtools::source_url("https://github.com/jjcurtin/lab_support/blob/main/fun_ml.R?raw=true")

theme_set(theme_classic())

path_data <- "survey_project/data"
```

2 Methods

2.1 Data Collection

The data used for this study were collected through Prolific, an online research platform that allows researchers to pay participants to complete surveys. The survey used for this study was created in Qualtrics, and then linked to the researcher's Prolific account. From there, the study was published and data collection began.

51 paarticipants were recruited in this study. The completed a 19-item sruvey which asked them about their Prolfic member ID (this was needed to be able to pay participants) and questions about lifestyle factors and well-being. The lifetsyle factor questions asked about factors such as sleep, nutrition, exercise, social connections, screen time, time spent in nature, and stress management. Survey questions were a good mix of categorical and numeric data types across all categories.

A data dictionary containing all variable names, survey items, data types, and units of measurement is

included below.

```
# data dictionary
data_dictionary <- tribble(
  ~variable_name,        ~question,

  "sleep",               "On average, how many hours of sleep do you get in one night?",
  "well_rested",         "I feel well-rested when I wake up.",
  "consistent",          "My sleeping/waking routine is consistent.",

  "produce",             "How many servings of fruits/vegetables do you eat daily?",
  "processed_food",      "I eat processed foods (fried, sugary, fast food).",
  "breakfast",           "Do you regularly eat breakfast?",

  "exercise",            "Minutes of physical activity per day?",
  "fav_activity",        "What type of physical activity is your favorite?",

  "belonging",           "I feel a sense of belonging in my everyday interactions.",
  "mean_conv",           "How many times a week do you have meaningful conversations with someone you lo

  "social_media",        "Total hours per day spent on social media?",
  "soc_med_prec",        "I put off tasks to spend time on social media.",

  "nature",              "Hours per week spent outdoors in a natural setting?",
  "refreshed",           "How often do you feel refreshed after being in nature?",
  "accessible",          "How accessible are natural spaces to you?",

  "under_pressure",      "How often do you feel like you're under pressure?",
  "manage_stress",       "I can manage my stress effectively when it arises.",

  "life_satisfaction",   "How satisfied are you with your life overall?",
  "well_being",          "How would you rate your current mental well-being?",
)

write_csv(data_dictionary, "data_dictionary.csv")
```

```
# read in data dictionary
read_csv("data_dictionary.csv", show_col_types = FALSE) %>%
  knitr::kable(caption = "Data Dictionary") %>%
  kableExtra::kable_styling(full_width = FALSE)
```

Table 1: Data Dictionary

variable_name	question	type
sleep	On average, how many hours of sleep do you get in one night?	Numeric
well_rested	I feel well-rested when I wake up.	Likert (1–5)
consistent	My sleeping/waking routine is consistent.	Likert (1–5)
produce	How many servings of fruits/vegetables do you eat daily?	Numeric
processed_food	I eat processed foods (fried, sugary, fast food).	Ordinal
breakfast	Do you regularly eat breakfast?	Binary
exercise	Minutes of physical activity per day?	Numeric
fav_activity	What type of physical activity is your favorite?	Free text
belonging	I feel a sense of belonging in my everyday interactions.	Likert (1–5)
mean_conv	How many times a week do you have meaningful conversations with someone you love?	Numeric

social_media	Total hours per day spent on social media?	Numeric
soc_med_prec	I put off tasks to spend time on social media.	Likert (1–5)
nature	Hours per week spent outdoors in a natural setting?	Numeric
refreshed	How often do you feel refreshed after being in nature?	Likert (1–5)
accessible	How accessible are natural spaces to you?	Likert (1–5)
under_pressure	How often do you feel like you're under pressure?	Likert (1–5)
manage_stress	I can manage my stress effectively when it arises.	Likert (1–5)
life_satisfaction	How satisfied are you with your life overall?	Numeric
well_being	How would you rate your current mental well-being?	Numeric

2.2 Data Cleaning

To begin this data analysis, two separate data files had to be loaded into R. One dataset (data1) was obtained from Prolific and contains demographic information of the particpants in this study. The second dataset (data2) was obtained from Qualtrics and contains the surveys responses, which is primarily the data of interest in this study.

```
# loading in datasets
data1 <- read_csv(here::here(path_data, "demographic_data.csv")) |>
  janitor::clean_names() |>
  glimpse()
```

```
Rows: 52
Columns: 21
$ submission_id                <chr> "6865526559fd16b818cb03c9", "6865526a83b~
$ participant_id               <chr> "657f5cb338e31904b66cd970", "66a02d45a7d~
$ status                       <chr> "APPROVED", "APPROVED", "APPROVED", "APP~
$ custom_study_tncs_accepted_at <chr> "Not Applicable", "Not Applicable", "Not~
$ started_at                   <dttm> 2025-07-02 15:38:17, 2025-07-02 15:38:2~
$ completed_at                 <dttm> 2025-07-02 15:39:51, 2025-07-02 15:40:1~
$ reviewed_at                  <dttm> 2025-07-02 21:09:37, 2025-07-02 21:09:3~
$ archived_at                  <dttm> 2025-07-02 15:39:52, 2025-07-02 15:40:1~
$ time_taken                   <dbl> 95, 112, 351, 169, 143, 112, 133, 105, 1~
$ completion_code              <chr> "C1DCX3LH", "C1DCX3LH", "C1DCX3LH", "C1D~
$ total_approvals              <dbl> 3590, 2856, 1424, 2116, 2252, 6350, 6261~
$ fluent_languages             <chr> "English", "English", "English", "Englis~
$ age                          <dbl> 47, 26, 31, 50, 60, 34, 37, 36, 33, 72, ~
$ sex                          <chr> "Female", "Male", "Male", "Male", "Femal~
$ ethnicity_simplified         <chr> "White", "White", "White", "White", "Whi~
$ country_of_birth             <chr> "United States", "United States", "Unite~
$ country_of_residence         <chr> "United States", "United States", "Unite~
$ nationality                  <chr> "United States", "United States", "Unite~
$ language                     <chr> "English", "English", "English", "Englis~
$ student_status               <chr> "DATA_EXPIRED", "Yes", "No", "DATA_EXPIR~
$ employment_status            <chr> "Not in paid work (e.g. homemaker', 'ret~
```

```
data2 <- read_csv(here::here(path_data, "755_survey_data.csv")) |>
  janitor::clean_names() |>
  glimpse()
```

```
Rows: 66
Columns: 37
$ start_date           <chr> "Start Date", "{\"ImportId\":\"startDate\",\"time~
$ end_date             <chr> "End Date", "{\"ImportId\":\"endDate\",\"timeZone~
$ status               <chr> "Response Type", "{\"ImportId\":\"status\"}", "1"~
```

```
$ ip_address           <chr> "IP Address", "{\"ImportId\":\"ipAddress\"}", NA,~
$ progress             <chr> "Progress", "{\"ImportId\":\"progress\"}", "100",~
$ duration_in_seconds  <chr> "Duration (in seconds)", "{\"ImportId\":\"duratio~
$ finished             <chr> "Finished", "{\"ImportId\":\"finished\"}", "1", "~
$ recorded_date        <chr> "Recorded Date", "{\"ImportId\":\"recordedDate\",~
$ response_id          <chr> "Response ID", "{\"ImportId\":\"_recordId\"}", "R~
$ recipient_last_name  <chr> "Recipient Last Name", "{\"ImportId\":\"recipient~
$ recipient_first_name <chr> "Recipient First Name", "{\"ImportId\":\"recipien~
$ recipient_email      <chr> "Recipient Email", "{\"ImportId\":\"recipientEmai~
$ external_reference   <chr> "External Data Reference", "{\"ImportId\":\"exter~
$ location_latitude    <chr> "Location Latitude", "{\"ImportId\":\"locationLat~
$ location_longitude   <chr> "Location Longitude", "{\"ImportId\":\"locationLo~
$ distribution_channel <chr> "Distribution Channel", "{\"ImportId\":\"distribu~
$ user_language        <chr> "User Language", "{\"ImportId\":\"userLanguage\"}~
$ q1                   <chr> "Please provide your Prolific ID.", "{\"ImportId\~
$ q2                   <chr> "On average, how many hours of sleep do you get i~
$ q3                   <chr> "I feel well-rested when I wake up.", "{\"ImportI~
$ q4                   <chr> "My sleeping/waking routine is consistent. (sleep~
$ q5                   <chr> "How many servings of fruits/vegetables do you ea~
$ q6                   <chr> "I eat processed foods (fried, sugary, fast food)~
$ q7                   <chr> "Do you regularly eat breakfast?", "{\"ImportId\"~
$ q8                   <chr> "Approximately how many minutes of physical activ~
$ q9                   <chr> "What type of physical activity is your favorite?~
$ q10                  <chr> "I feel a sense of belonging in my everyday inter~
$ q11                  <chr> "How many times in a week do you have meaningful ~
$ q12                  <chr> "How many total hours do you spend per day on soc~
$ q13                  <chr> "I spend time on social media, putting off tasks ~
$ q20                  <chr> "On average, how many hours per week do you spend~
$ q24                  <chr> "How often do you feel refreshed or recharged aft~
$ q25                  <chr> "How accessible are natural spaces (parks, trails~
$ q14                  <chr> "How often do you feel like you're under pressure~
$ q15                  <chr> "I can manage my stress effectively when it arise~
$ q18                  <chr> "On a scale from 0 to 10, how satisfied are you w~
$ q19                  <chr> "On a scale from 0 to 10, how would you rate your~
```

After loading in the datasets, they had to be merged together to be usable in this analysis. The datasets were merged on the "ID" column, which is the only identical column across both datasets. Doing this step also removes the uneccessary observations in data2, which were created in the trialing and testing of the study survey.

```
# preparing to merge data1 on the "id" column
data1 <- data1 %>%
  rename(id = participant_id)
```

```
# preparing to merge data2 on the "id" column
data2 <- data2 %>%
  rename(id = q1)
```

```
# merging qualtrics & prolfic data
data <- merge(data1, data2, by="id") %>% glimpse()
```

```
Rows: 51
Columns: 57
$ id                          <chr> "5591cd21fdf99b63126f6c60", "58d552e09db~
$ submission_id               <chr> "6865534ca31b91f13b545c5b", "68655346e29~
$ status.x                    <chr> "APPROVED", "APPROVED", "APPROVED", "APP~
```

```
$ custom_study_tncs_accepted_at <chr> "Not Applicable", "Not Applicable", "Not~
$ started_at                    <dttm> 2025-07-02 15:42:08, 2025-07-02 15:42:0~
$ completed_at                  <dttm> 2025-07-02 15:44:53, 2025-07-02 15:45:0~
$ reviewed_at                   <dttm> 2025-07-02 21:09:49, 2025-07-02 21:09:4~
$ archived_at                   <dttm> 2025-07-02 15:44:53, 2025-07-02 15:45:0~
$ time_taken                    <dbl> 165, 179, 120, 256, 253, 133, 351, 105, ~
$ completion_code               <chr> "C1DCX3LH", "C1DCX3LH", "C1DCX3LH", "C1D~
$ total_approvals               <dbl> 8812, 5046, 1403, 9915, 6342, 6261, 1424~
$ fluent_languages              <chr> "English", "English", "English", "Englis~
$ age                           <dbl> 34, 48, 37, 60, 44, 37, 31, 36, 35, 52, ~
$ sex                           <chr> "Male", "Female", "Female", "Male", "Fem~
$ ethnicity_simplified          <chr> "White", "White", "White", "White", "Asi~
$ country_of_birth              <chr> "United States", "United States", "Unite~
$ country_of_residence          <chr> "United States", "United States", "Unite~
$ nationality                   <chr> "United States", "United States", "Unite~
$ language                      <chr> "English", "English", "English", "Englis~
$ student_status                <chr> "No", "No", "No", "DATA_EXPIRED", "DATA_~
$ employment_status             <chr> "Full-Time", "Full-Time", "Not in paid w~
$ start_date                    <chr> "2025-07-02 09:42:10", "2025-07-02 09:42~
$ end_date                      <chr> "2025-07-02 09:43:23", "2025-07-02 09:44~
$ status.y                      <chr> "0", "0", "0", "0", "0", "0", "0", "0", ~
$ ip_address                    <chr> "174.238.160.216", "172.59.122.36", "70.~
$ progress                      <chr> "100", "100", "100", "100", "100", "100"~
$ duration_in_seconds           <chr> "72", "156", "100", "73", "99", "114", "~
$ finished                      <chr> "1", "1", "1", "1", "1", "1", "1", "1", ~
$ recorded_date                 <chr> "2025-07-02 09:43:23", "2025-07-02 09:44~
$ response_id                   <chr> "R_5KqbNdNZBWEIWrl", "R_1UVCjpnFd0ZCdGJ"~
$ recipient_last_name           <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ recipient_first_name          <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ recipient_email               <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ external_reference            <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ location_latitude             <chr> "35.8323", "35.4662", "33.3351", "35.546~
$ location_longitude            <chr> "-83.9451", "-97.5168", "-111.7375", "-7~
$ distribution_channel          <chr> "anonymous", "anonymous", "anonymous", "~
$ user_language                 <chr> "EN", "EN", "EN", "EN", "EN", "EN", "EN"~
$ q2                            <chr> "6", "7", "5", "8", "7", "7", "6", "7", ~
$ q3                            <chr> "4", "4", "1", "5", "4", "4", "4", "5", ~
$ q4                            <chr> "5", "4", "1", "5", "5", "5", "5", "5", ~
$ q5                            <chr> "1", "3", "6", "4", "2", "5", "0", "2", ~
$ q6                            <chr> "2", "4", "4", "2", "3", "3", "2", "1", ~
$ q7                            <chr> "1", "2", "2", "2", "1", "2", "2", "1", ~
$ q8                            <chr> "30", "30", "30", "120", "30", "35", "30~
$ q9                            <chr> "Walking", "walking", "Swimming", "walki~
$ q10                           <chr> "1", "1", "2", "1", "1", "2", "4", "1", ~
$ q11                           <chr> "10", "10", "4", "7", "3", "2", "1", "4"~
$ q12                           <chr> "3", "1", "2", "0", "1", "1", "0", "60",~
$ q13                           <chr> "2", "1", "2", "1", "3", "2", "1", "1", ~
$ q20                           <chr> "0", "1", "3", "30", "1", "3", "0", "30"~
$ q24                           <chr> "2", "2", "3", "5", "4", "3", "2", "5", ~
$ q25                           <chr> "7", "6", "6", "8", "7", "6", "8", "5", ~
$ q14                           <chr> "4", "2", "2", "2", "3", "3", "5", "2", ~
$ q15                           <chr> "5", "4", "4", "5", "3", "5", "2", "5", ~
$ q18                           <chr> "7", "8", "5", "10", "8", "7", "1", "8",~
$ q19                           <chr> "9", "9", "5", "10", "8", "7", "1", "10"~
```

After merging both dataset into one (data), the survey questions were renamed. The new names ensure ease in identifying the questions quickly during analysis, streamlining the coding process.

```
# renaming qualtrics questions
data <- data %>%
  rename(sleep = q2) %>%
  rename(well_rested = q3) %>%
  rename(consistent = q4) %>%
  rename(produce = q5) %>%
  rename(processed_food = q6) %>%
  rename(breakfast = q7) %>%
  rename(exercise = q8) %>%
  rename(fav_activity = q9) %>%
  rename(belonging = q10) %>%
  rename(mean_conv = q11) %>%
  rename(social_media = q12) %>%
  rename(soc_med_prec = q13) %>%
  rename(nature = q20) %>%
  rename(refreshed = q24) %>%
  rename(accessible = q25) %>%
  rename(under_pressure = q14) %>%
  rename(manage_stress = q15) %>%
  rename(life_satisfaction = q18) %>%
  rename(well_being = q19)
```

The next step involved removing unnecessary columns, which were a part of the datasets downloaded from Qualtric and Prolific. These are columns that will most definitiely not be used in this analysis. This step ensures a smaller and more workable dataset.

```
# removing columns I'll never need
data <- data %>%
  select(-archived_at, -completion_code, -country_of_residence, -language,
         -start_date, -end_date, -status.y, -ip_address, -progress, -finished,
         -recorded_date, -response_id, -recipient_email, -recipient_first_name,
         -recipient_last_name, -external_reference, -distribution_channel,
         -user_language, -submission_id, -status.x, -custom_study_tncs_accepted_at,
         -started_at, -completed_at, -reviewed_at)
```

2.3 Exploratory Data Analysis

To begin exploratory data analysis, the dataset was briefly summarized using skim(). This dataset contains 51 observations with 33 variables. #0 variables were loaded in as characters and likely need to be changed, and 3 are numeric. This dataset contains no missing values. Minimum and maximum values for each column were also explored.

```
# brief scan of data
skim(data)
```

Table 2: Data summary

Name	data
Number of rows	51
Number of columns	33
Column type frequency:	
character	30

numeric	3
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
id	0	1	24	24	0	51	0
fluent_languages	0	1	7	25	0	5	0
sex	0	1	4	6	0	2	0
ethnicity_simplified	0	1	5	5	0	5	0
country_of_birth	0	1	5	13	0	3	0
nationality	0	1	13	13	0	1	0
student_status	0	1	2	12	0	3	0
employment_status	0	1	9	56	0	5	0
duration_in_seconds	0	1	2	3	0	49	0
location_latitude	0	1	5	7	0	48	0
location_longitude	0	1	7	9	0	48	0
sleep	0	1	1	3	0	6	0
well_rested	0	1	1	1	0	5	0
consistent	0	1	1	1	0	5	0
produce	0	1	1	3	0	8	0
processed_food	0	1	1	1	0	5	0
breakfast	0	1	1	1	0	2	0
exercise	0	1	1	8	0	14	0
fav_activity	0	1	5	192	0	33	0
belonging	0	1	1	1	0	5	0
mean_conv	0	1	1	2	0	14	0
social_media	0	1	1	2	0	8	0
soc_med_prec	0	1	1	1	0	4	0
nature	0	1	1	2	0	15	0
refreshed	0	1	1	1	0	5	0
accessible	0	1	1	1	0	5	0
under_pressure	0	1	1	1	0	5	0
manage_stress	0	1	1	1	0	5	0
life_satisfaction	0	1	1	2	0	9	0
well_being	0	1	1	2	0	10	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
time_taken	0	1	192.04	109.52	70	122.5	154	230.5	546	
total_approvals	0	1	3602.04	2805.66	166	1454.5	2756	5120.5	12381	
age	0	1	41.53	13.81	21	31.0	39	51.0	72	

```
glimpse(data)
```

```
Rows: 51
Columns: 33
$ id                   <chr> "5591cd21fdf99b63126f6c60", "58d552e09dbe3f000125~
```

```
$ time_taken           <dbl> 165, 179, 120, 256, 253, 133, 351, 105, 111, 100,~
$ total_approvals      <dbl> 8812, 5046, 1403, 9915, 6342, 6261, 1424, 8850, 4~
$ fluent_languages     <chr> "English", "English", "English", "English", "Indo~
$ age                  <dbl> 34, 48, 37, 60, 44, 37, 31, 36, 35, 52, 32, 31, 2~
$ sex                  <chr> "Male", "Female", "Female", "Male", "Female", "Ma~
$ ethnicity_simplified <chr> "White", "White", "White", "White", "Asian", "Asi~
$ country_of_birth     <chr> "United States", "United States", "United States"~
$ nationality          <chr> "United States", "United States", "United States"~
$ student_status       <chr> "No", "No", "No", "DATA_EXPIRED", "DATA_EXPIRED",~
$ employment_status    <chr> "Full-Time", "Full-Time", "Not in paid work (e.g.~
$ duration_in_seconds  <chr> "72", "156", "100", "73", "99", "114", "150", "91~
$ location_latitude    <chr> "35.8323", "35.4662", "33.3351", "35.5466", "34.0~
$ location_longitude   <chr> "-83.9451", "-97.5168", "-111.7375", "-77.0522", ~
$ sleep                <chr> "6", "7", "5", "8", "7", "7", "6", "7", "8", "5",~
$ well_rested          <chr> "4", "4", "1", "5", "4", "4", "4", "5", "4", "2",~
$ consistent           <chr> "5", "4", "1", "5", "5", "5", "5", "5", "4", "4",~
$ produce              <chr> "1", "3", "6", "4", "2", "5", "0", "2", "6", "1",~
$ processed_food       <chr> "2", "4", "4", "2", "3", "3", "2", "1", "2", "5",~
$ breakfast            <chr> "1", "2", "2", "2", "1", "2", "2", "1", "2", "1",~
$ exercise             <chr> "30", "30", "30", "120", "30", "35", "30", "60", ~
$ fav_activity         <chr> "Walking", "walking", "Swimming", "walking", "Run~
$ belonging            <chr> "1", "1", "2", "1", "1", "2", "4", "1", "4", "5",~
$ mean_conv            <chr> "10", "10", "4", "7", "3", "2", "1", "4", "2", "2~
$ social_media         <chr> "3", "1", "2", "0", "1", "1", "0", "60", "7", "0"~
$ soc_med_prec         <chr> "2", "1", "2", "1", "3", "2", "1", "1", "1", "1",~
$ nature               <chr> "0", "1", "3", "30", "1", "3", "0", "30", "4", "5~
$ refreshed            <chr> "2", "2", "3", "5", "4", "3", "2", "5", "1", "4",~
$ accessible           <chr> "7", "6", "6", "8", "7", "6", "8", "5", "7", "8",~
$ under_pressure       <chr> "4", "2", "2", "2", "3", "3", "5", "2", "2", "4",~
$ manage_stress        <chr> "5", "4", "4", "5", "3", "5", "2", "5", "2", "3",~
$ life_satisfaction    <chr> "7", "8", "5", "10", "8", "7", "1", "8", "3", "1"~
$ well_being           <chr> "9", "9", "5", "10", "8", "7", "1", "10", "3", "1~
```

Character variables were converted to factors for later use. In the process of converting variable classes, one error was noted in the dataset. One participant entered "one hour" in the exercise column, when they should have entered "60 minutes". This error prevents the column from being able to convert to a numeric class, the the response for this observation was manually changed to "60".

```
# converting character variables to factor
data <- data %>%
  mutate(across(where(is.character), as.factor)) %>%
  glimpse()
```

```
Rows: 51
Columns: 33
$ id                   <fct> 5591cd21fdf99b63126f6c60, 58d552e09dbe3f00012591f~
$ time_taken           <dbl> 165, 179, 120, 256, 253, 133, 351, 105, 111, 100,~
$ total_approvals      <dbl> 8812, 5046, 1403, 9915, 6342, 6261, 1424, 8850, 4~
$ fluent_languages     <fct> "English", "English", "English", "English", "Indo~
$ age                  <dbl> 34, 48, 37, 60, 44, 37, 31, 36, 35, 52, 32, 31, 2~
$ sex                  <fct> Male, Female, Female, Male, Female, Male, Male, M~
$ ethnicity_simplified <fct> White, White, White, White, Asian, Asian, White, ~
$ country_of_birth     <fct> United States, United States, United States, Unit~
$ nationality          <fct> United States, United States, United States, Unit~
$ student_status       <fct> No, No, No, DATA_EXPIRED, DATA_EXPIRED, No, No, D~
```

```
$ employment_status    <fct> "Full-Time", "Full-Time", "Not in paid work (e.g.~
$ duration_in_seconds  <fct> 72, 156, 100, 73, 99, 114, 150, 91, 76, 83, 92, 1~
$ location_latitude    <fct> 35.8323, 35.4662, 33.3351, 35.5466, 34.0515, 44.8~
$ location_longitude   <fct> -83.9451, -97.5168, -111.7375, -77.0522, -84.0713~
$ sleep                <fct> 6, 7, 5, 8, 7, 7, 6, 7, 8, 5, 8, 7, 7, 5, 6, 7, 7~
$ well_rested          <fct> 4, 4, 1, 5, 4, 4, 4, 5, 4, 2, 5, 2, 4, 2, 2, 2, 1~
$ consistent           <fct> 5, 4, 1, 5, 5, 5, 5, 5, 4, 4, 1, 3, 4, 1, 4, 4, 4~
$ produce              <fct> 1, 3, 6, 4, 2, 5, 0, 2, 6, 1, 1, 1, 3, 3, 1, 2, 2~
$ processed_food       <fct> 2, 4, 4, 2, 3, 3, 2, 1, 2, 5, 2, 2, 3, 2, 2, 3, 2~
$ breakfast            <fct> 1, 2, 2, 2, 1, 2, 2, 1, 2, 1, 2, 2, 1, 2, 2, 2, 2~
$ exercise             <fct> 30, 30, 30, 120, 30, 35, 30, 60, 60, 0, 30, 30, 3~
$ fav_activity         <fct> "Walking", "walking", "Swimming", "walking", "Run~
$ belonging            <fct> 1, 1, 2, 1, 1, 2, 4, 1, 4, 5, 3, 2, 2, 5, 4, 2, 5~
$ mean_conv            <fct> 10, 10, 4, 7, 3, 2, 1, 4, 2, 2, 1, 3, 10, 12, 1, ~
$ social_media         <fct> 3, 1, 2, 0, 1, 1, 0, 60, 7, 0, 0, 2, 4, 1, 2, 1, ~
$ soc_med_prec         <fct> 2, 1, 2, 1, 3, 2, 1, 1, 1, 1, 1, 2, 2, 1, 1, 2, 4~
$ nature               <fct> 0, 1, 3, 30, 1, 3, 0, 30, 4, 5, 2, 1, 2, 12, 1, 1~
$ refreshed            <fct> 2, 2, 3, 5, 4, 3, 2, 5, 1, 4, 2, 5, 2, 1, 4, 4, 5~
$ accessible           <fct> 7, 6, 6, 8, 7, 6, 8, 5, 7, 8, 7, 5, 6, 5, 8, 7, 7~
$ under_pressure       <fct> 4, 2, 2, 2, 3, 3, 5, 2, 2, 4, 1, 2, 2, 5, 4, 2, 2~
$ manage_stress        <fct> 5, 4, 4, 5, 3, 5, 2, 5, 2, 3, 5, 4, 2, 1, 5, 4, 2~
$ life_satisfaction    <fct> 7, 8, 5, 10, 8, 7, 1, 8, 3, 1, 7, 5, 6, 1, 4, 6, ~
$ well_being           <fct> 9, 9, 5, 10, 8, 7, 1, 10, 3, 1, 7, 5, 6, 1, 3, 7,~
```

```r
# cleaning responses in "exercise" column
data <- data %>%
  mutate(exercise = case_when(
    exercise == "one hour" ~ "60",
    TRUE ~ exercise
  ) %>% parse_number())
```

With the response error corrected, the necessary numeric columns were converted into the correct class.

```r
# converting numeric variables
data <- data %>%
  mutate(across(c(sleep, produce, exercise, mean_conv, social_media, nature,
                  life_satisfaction, well_being, duration_in_seconds),
                ~ as.numeric(as.character(.)))) %>%
  glimpse()
```

```
Rows: 51
Columns: 33
$ id                   <fct> 5591cd21fdf99b63126f6c60, 58d552e09dbe3f00012591f~
$ time_taken           <dbl> 165, 179, 120, 256, 253, 133, 351, 105, 111, 100,~
$ total_approvals      <dbl> 8812, 5046, 1403, 9915, 6342, 6261, 1424, 8850, 4~
$ fluent_languages     <fct> "English", "English", "English", "English", "Indo~
$ age                  <dbl> 34, 48, 37, 60, 44, 37, 31, 36, 35, 52, 32, 31, 2~
$ sex                  <fct> Male, Female, Female, Male, Female, Male, Male, M~
$ ethnicity_simplified <fct> White, White, White, White, Asian, Asian, White, ~
$ country_of_birth     <fct> United States, United States, United States, Unit~
$ nationality          <fct> United States, United States, United States, Unit~
$ student_status       <fct> No, No, No, DATA_EXPIRED, DATA_EXPIRED, No, No, D~
$ employment_status    <fct> "Full-Time", "Full-Time", "Not in paid work (e.g.~
$ duration_in_seconds  <dbl> 72, 156, 100, 73, 99, 114, 150, 91, 76, 83, 92, 1~
$ location_latitude    <fct> 35.8323, 35.4662, 33.3351, 35.5466, 34.0515, 44.8~
```

```
$ location_longitude   <fct> -83.9451, -97.5168, -111.7375, -77.0522, -84.0713~
$ sleep                <dbl> 6.0, 7.0, 5.0, 8.0, 7.0, 7.0, 6.0, 7.0, 8.0, 5.0,~
$ well_rested          <fct> 4, 4, 1, 5, 4, 4, 4, 5, 4, 2, 5, 2, 4, 2, 2, 2, 1~
$ consistent           <fct> 5, 4, 1, 5, 5, 5, 5, 5, 4, 4, 1, 3, 4, 1, 4, 4, 4~
$ produce              <dbl> 1.0, 3.0, 6.0, 4.0, 2.0, 5.0, 0.0, 2.0, 6.0, 1.0,~
$ processed_food       <fct> 2, 4, 4, 2, 3, 3, 2, 1, 2, 5, 2, 2, 3, 2, 2, 3, 2~
$ breakfast            <fct> 1, 2, 2, 2, 1, 2, 2, 1, 2, 1, 2, 2, 1, 2, 2, 2, 2~
$ exercise             <dbl> 30, 30, 30, 120, 30, 35, 30, 60, 60, 0, 30, 30, 3~
$ fav_activity         <fct> "Walking", "walking", "Swimming", "walking", "Run~
$ belonging            <fct> 1, 1, 2, 1, 1, 2, 4, 1, 4, 5, 3, 2, 2, 5, 4, 2, 5~
$ mean_conv            <dbl> 10, 10, 4, 7, 3, 2, 1, 4, 2, 2, 1, 3, 10, 12, 1, ~
$ social_media         <dbl> 3, 1, 2, 0, 1, 1, 0, 60, 7, 0, 0, 2, 4, 1, 2, 1, ~
$ soc_med_prec         <fct> 2, 1, 2, 1, 3, 2, 1, 1, 1, 1, 1, 2, 2, 1, 1, 2, 4~
$ nature               <dbl> 0, 1, 3, 30, 1, 3, 0, 30, 4, 5, 2, 1, 2, 12, 1, 1~
$ refreshed            <fct> 2, 2, 3, 5, 4, 3, 2, 5, 1, 4, 2, 5, 2, 1, 4, 4, 5~
$ accessible           <fct> 7, 6, 6, 8, 7, 6, 8, 5, 7, 8, 7, 5, 6, 5, 8, 7, 7~
$ under_pressure       <fct> 4, 2, 2, 2, 3, 3, 5, 2, 2, 4, 1, 2, 2, 5, 4, 2, 2~
$ manage_stress        <fct> 5, 4, 4, 5, 3, 5, 2, 5, 2, 3, 5, 4, 2, 1, 5, 4, 2~
$ life_satisfaction    <dbl> 7, 8, 5, 10, 8, 7, 1, 8, 3, 1, 7, 5, 6, 1, 4, 6, ~
$ well_being           <dbl> 9, 9, 5, 10, 8, 7, 1, 10, 3, 1, 7, 5, 6, 1, 3, 7,~
```

Additionally, one observation in the social_media column was manually altered. In this instance, it is believed that one participant accidentally entered "60 minutes" in their survey when they menat to enter "1 hour". Essentially, one participant responded in the wrong unit of measurement. This error was noticed later during EDA (during univariate analysis), but was fixed here in the script.

```
# cleaning responses in "social_media" column
data <- data %>%
  mutate(social_media = case_when(
    social_media == "60" ~ 1,
    TRUE ~ social_media
  ))
```

After summarizing the data one more time with skim(), 21 variables are now factors and 12 variables are numeric. The mean and percentile ranges of numeric variables (0%, 25%, 50%, 75%, & 100%), can now be explored.

```
# checking data again
skim(data)
```

Table 5: Data summary

Name	data
Number of rows	51
Number of columns	33
Column type frequency:	
factor	21
numeric	12
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
id	0	1	FALSE	51	559: 1, 58d: 1, 599: 1, 59e: 1
fluent_languages	0	1	FALSE	5	Eng: 47, Cro: 1, Eng: 1, Ind: 1
sex	0	1	FALSE	2	Mal: 33, Fem: 18
ethnicity_simplified	0	1	FALSE	5	Whi: 38, Asi: 6, Mix: 3, Oth: 3
country_of_birth	0	1	FALSE	3	Uni: 49, Ind: 1, Jap: 1
nationality	0	1	FALSE	1	Uni: 51
student_status	0	1	FALSE	3	No: 29, DAT: 16, Yes: 6
employment_status	0	1	FALSE	5	Ful: 21, DAT: 13, Not: 7, Par: 6
location_latitude	0	1	FALSE	48	37.: 4, 25.: 1, 26.: 1, 27.: 1
location_longitude	0	1	FALSE	48	-97: 4, -10: 1, -11: 1, -11: 1
well_rested	0	1	FALSE	5	4: 20, 2: 14, 1: 7, 5: 7
consistent	0	1	FALSE	5	4: 25, 5: 14, 2: 6, 1: 5
processed_food	0	1	FALSE	5	2: 24, 3: 15, 4: 5, 5: 4
breakfast	0	1	FALSE	2	2: 31, 1: 20
fav_activity	0	1	FALSE	33	wal: 5, Wal: 5, Swi: 4, Run: 3
belonging	0	1	FALSE	5	2: 21, 1: 11, 4: 9, 3: 5
soc_med_prec	0	1	FALSE	4	2: 22, 1: 18, 3: 6, 4: 5
refreshed	0	1	FALSE	5	4: 22, 2: 9, 3: 8, 5: 8
accessible	0	1	FALSE	5	7: 18, 6: 15, 8: 9, 5: 8
under_pressure	0	1	FALSE	5	2: 26, 4: 13, 3: 6, 1: 3
manage_stress	0	1	FALSE	5	4: 23, 5: 10, 2: 9, 3: 5

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
time_taken	0	1	192.04	109.52	70	122.5	154	230.5	546	
total_approvals	0	1	3602.04	2805.66	166	1454.5	2756	5120.5	12381	
age	0	1	41.53	13.81	21	31.0	39	51.0	72	
duration_in_seconds	0	1	155.20	100.27	59	98.0	124	168.0	508	
sleep	0	1	6.87	1.12	5	6.0	7	7.5	11	
produce	0	1	2.23	1.51	0	1.0	2	3.0	6	
exercise	0	1	42.25	39.74	0	20.0	30	60.0	180	
mean_conv	0	1	5.94	6.82	0	2.0	4	7.0	40	
social_media	0	1	2.10	1.57	0	1.0	2	3.0	7	
nature	0	1	5.70	6.75	0	1.5	4	7.0	30	
life_satisfaction	0	1	5.86	2.60	1	4.0	6	8.0	10	
well_being	0	1	6.20	2.70	1	4.5	7	8.0	10	

2.4 Univariate EDA

To begin univariate EDA, factor variables contained within the survey items are all grouped together and named "factor_vars".

```
# grouping survey factor variables
factor_vars <- c("well_rested", "consistent", "processed_food", "breakfast",
                 "belonging", "refreshed", "soc_med_prec", "accessible",
                 "under_pressure", "manage_stress")
```

Next, all demographic variables are grouped together and named "demographic_vars".

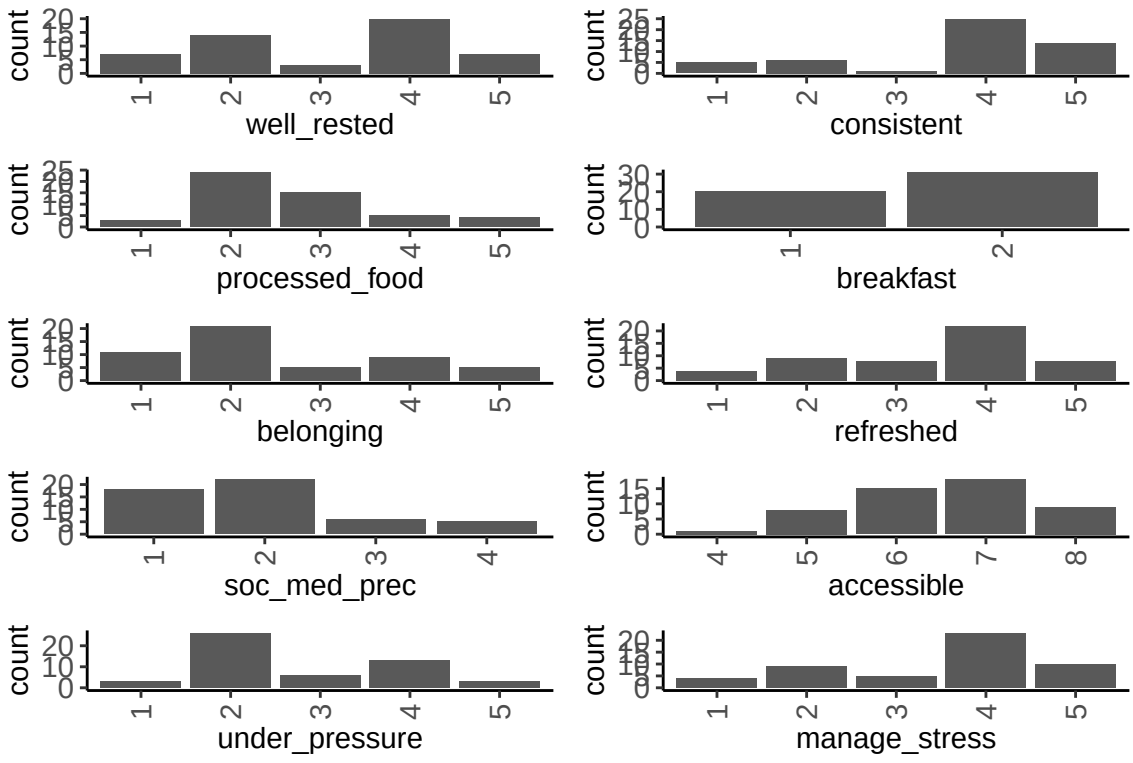
```
# grouping demographic variables
demographic_vars <- c("sex", "ethnicity_simplified",
                      "country_of_birth", "employment_status")
```

Then, all factor variables (survey items & demographic measures) are grouped together and named "all_factor_vars".

```
# grouping all factor variables
all_factor_vars <- c("sex", "ethnicity_simplified", "country_of_birth",
                     "employment_status", "well_rested", "consistent",
                     "soc_med_prec", "accessible", "under_pressure", "manage_stress")
```

The map() function is used to create bar plots of all variables in "factor_vars". The variable "well_rested" appears to have a clear bimodal distribution in feeling well-rested upon waking. "Consistent" has a slightly bimodal distribution, but mainly skews to the left towards having a consistent sleep/waking routine. "Refreshed", has a left skew, meaning more people feel refreshed or recharged after being in nature. "Accessible", also skew to the left, meaning more people find nature spaces (e.g. parks, trail, green areas, etc) very accessible. The univariate graph for "belonging" has a right skew, but this item was reverse coded on the survey. More poeple feel a sense of belonging in their day-to-day lives than not. "Soc_med_prec" also skews to the right, indicating that fewer people put off necessary tasks by spending time on social media than those that do not. The mode for "processed food" is eating processed food 1-2 days a week. More than half of the participants in this study eat breakfast every day.

```
# bar plots of categorical variables
data |>
  select(factor_vars) |>
  names() |>
  map(\(name) plot_bar(df = data, x = name)) |>
  plot_grid(plotlist = _, ncol = 2)
```



Responses for some of the variables in the demographic questions are long and ruin the formatting of bar

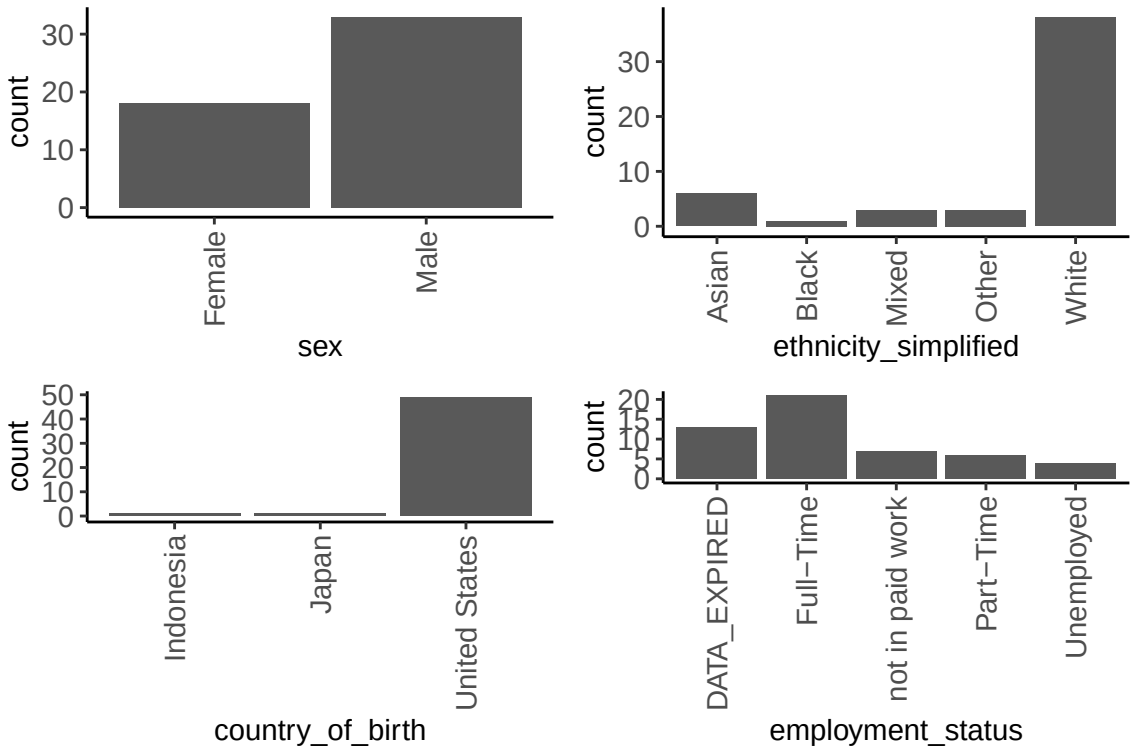
plots. Here, these responses are recoded and shortened to make bar plots more readable.

```
# shortening 'employment_status' responses
data <- data %>%
  mutate(employment_status = recode(
    employment_status,
    "Not in paid work (e.g. homemaker', 'retired or disabled)" = "not in paid work"
  ))

data <- data %>%
  mutate(employment_status = recode(
    employment_status,
    "Unemployed (and job seeking)" = "Unemployed"
  ))
```

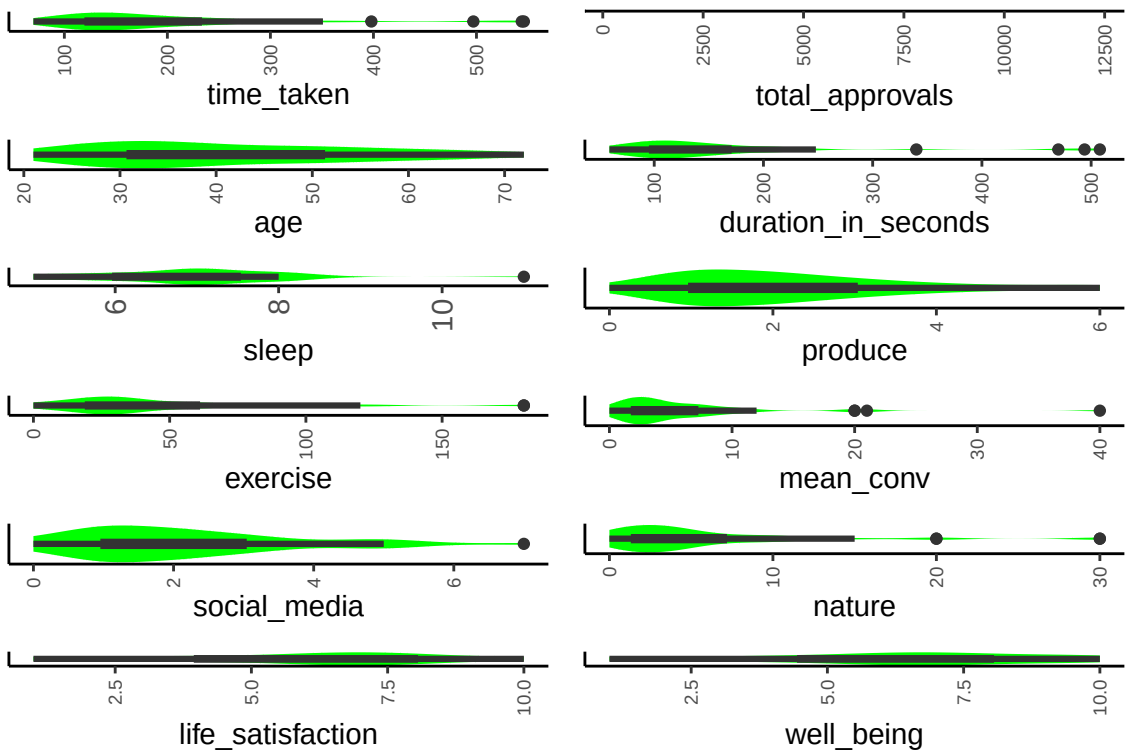
Below, bar plots are created using the map() function, showing all of the columns in "demographic_vars". The majority of this sample is White, male, and born in the United States. About half of the sample is employed full time and about one quarter of this sample have expired data for the smployment status question. This is essentially missing data that was not caught by the skim() function.

```
# bar plots of demographic variables
data |>
  select(demographic_vars) |>
  names() |>
  map(\(name) plot_bar(df = data, x = name)) |>
  plot_grid(plotlist = _, ncol = 2)
```



Next, violin plots are created of all numeric variables. These show some skewness and some outliers, but nothing concerning. Initially, this is where the response error in unit of measurement was discovered, as there was a very clear outlier. "Mean_conv" has an outlying observation, at 40 meaningful conversations had per week. This value will be left as is.

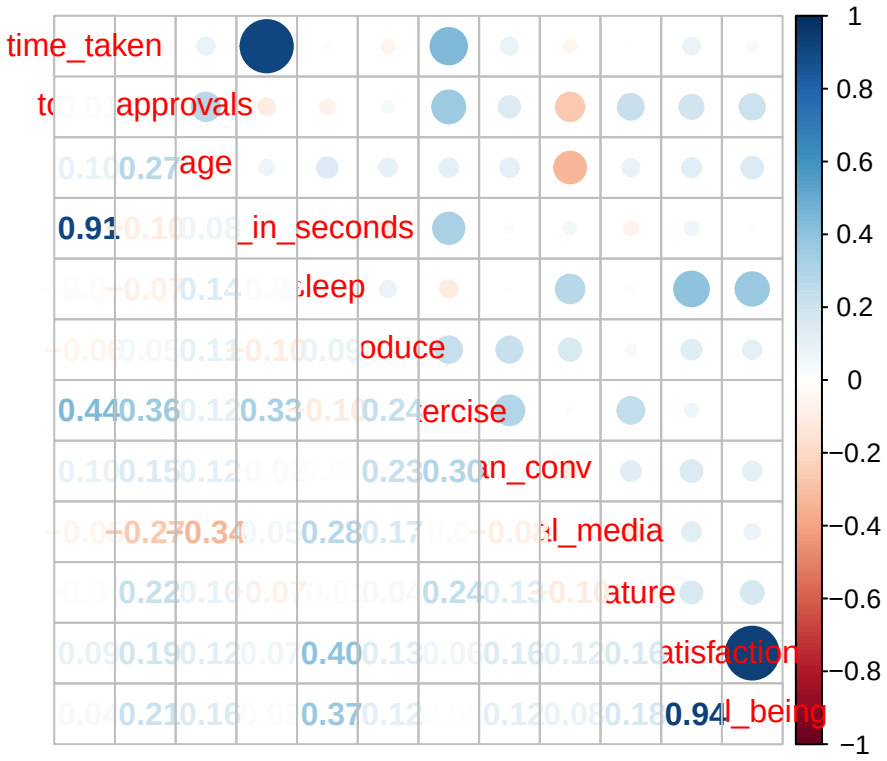
```
# violin plots of numeric variables
data |>
  select(where(is.numeric)) |>
  names() |>
  map(\(name) plot_box_violin(df = data, x = name)) |>
  plot_grid(plotlist = _, ncol = 2)
```



2.5 Bivariate EDA

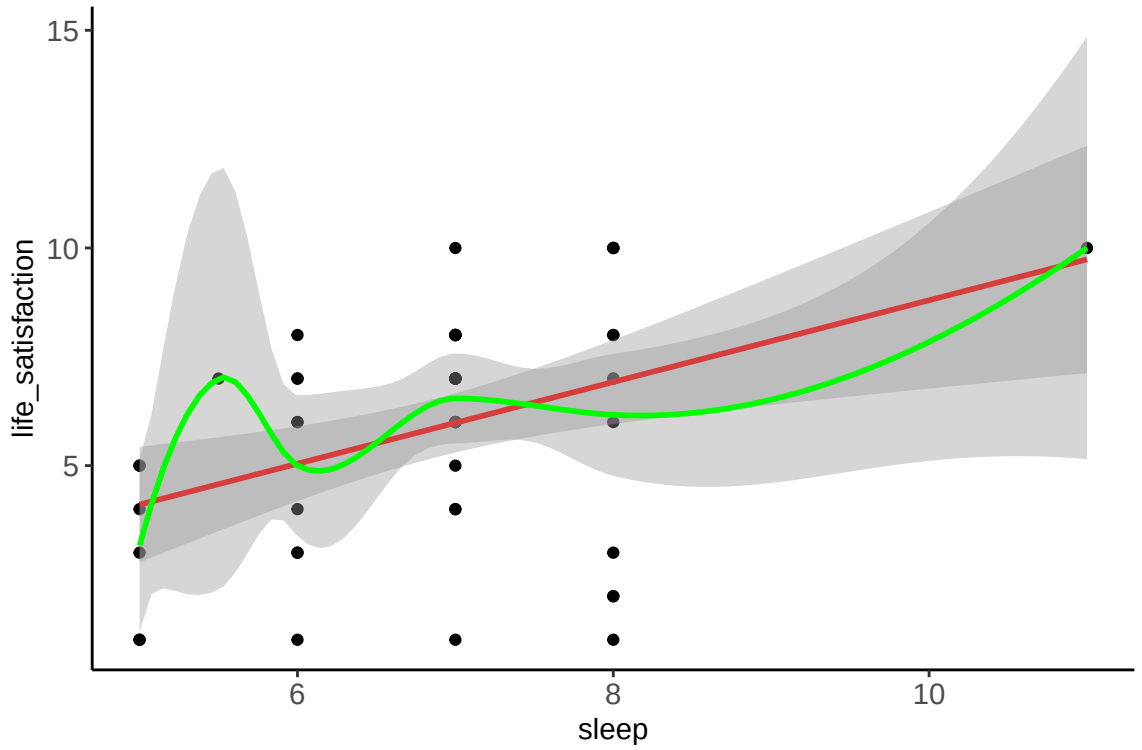
To begin bivariate analysis, a correlation matrix was created using all numeric variables from this dataset. This matrix shows a near perfect positive correlation between life satisfaction and well-being, indicating that as a dependent variable, both variables would perform similarly. Aside from that, there is a weak to moderate positive correlation between sleep and well-being worth noting.

```
# correlation matrix w/ numeric variables
data |>
  select(where(is.numeric)) |>
  cor(use = "pairwise.complete.obs") |>
  corrplot::corrplot.mixed()
```

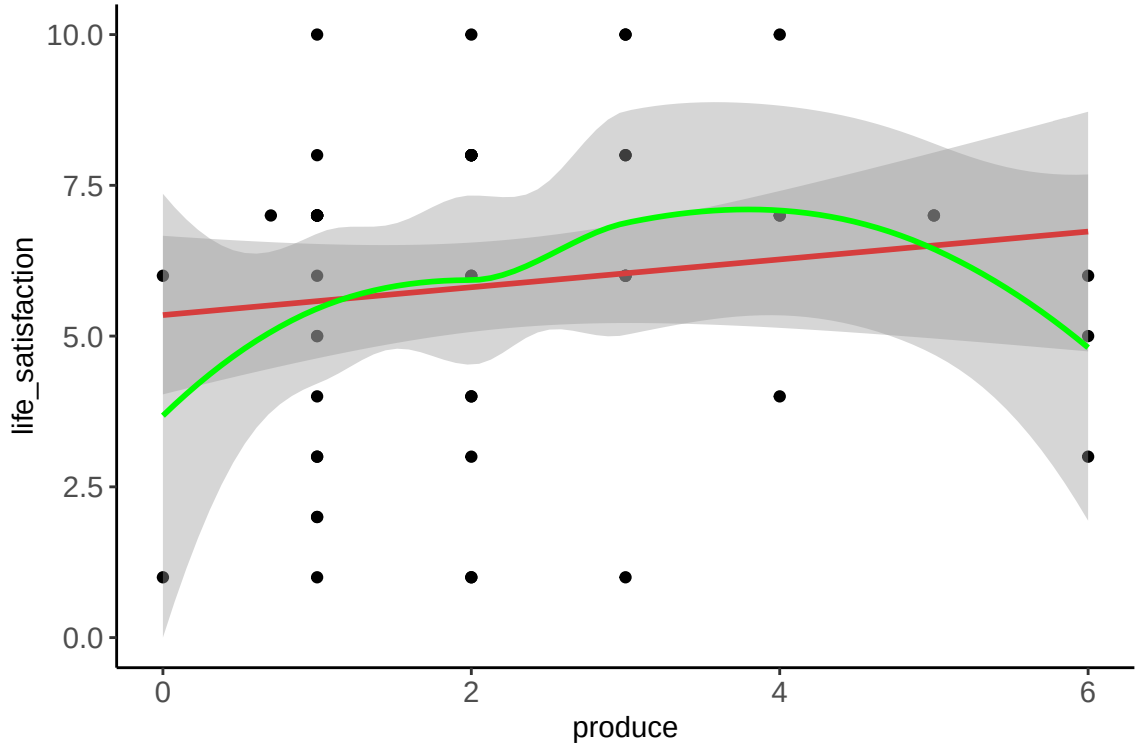


Next, scatter plots were created comparing life_satisfaction responses with numeric independent variables. Sleep shows a fairly consistent positive correlation with Life satisfaction, indicating that more hours of sleep per night may be correlated with greater life satisfaction. Time spent in nature produces an interesting graph; there appears to be a positive correlation between time spent in nature and life satisfaction. However, the points are very spread out on the graph, so this correlation may not be statistically significant. Fruit/vegetable consumption and age produce a parabolic shape when plotted against life satisfaction, which could be worth exploring.

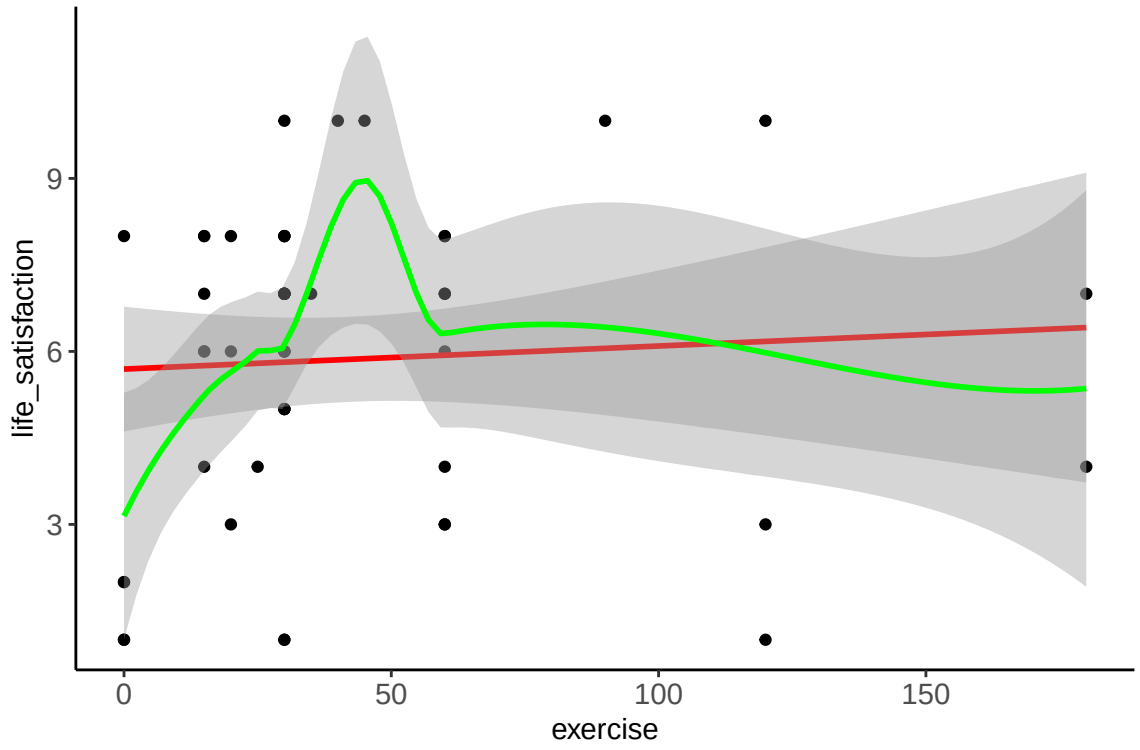
```
# scatter plots of independent variables against 'life_satisfaction'
data |> plot_scatter("sleep", "life_satisfaction")
```



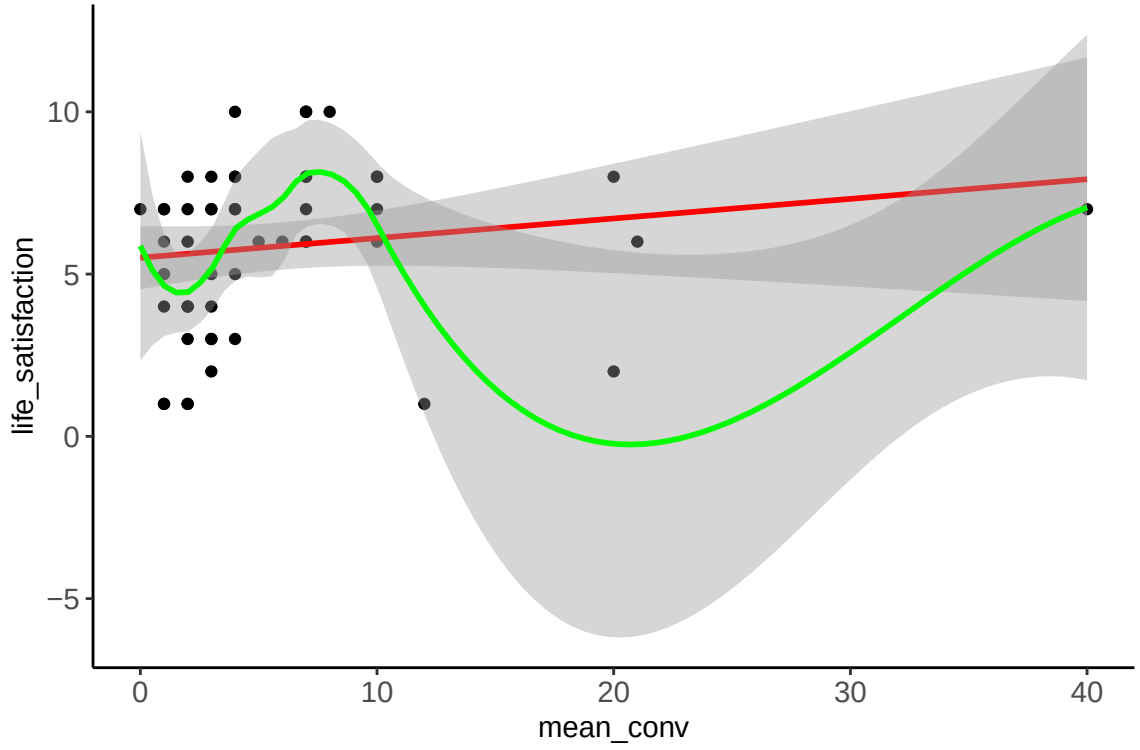

```
data |> plot_scatter("produce", "life_satisfaction")
```



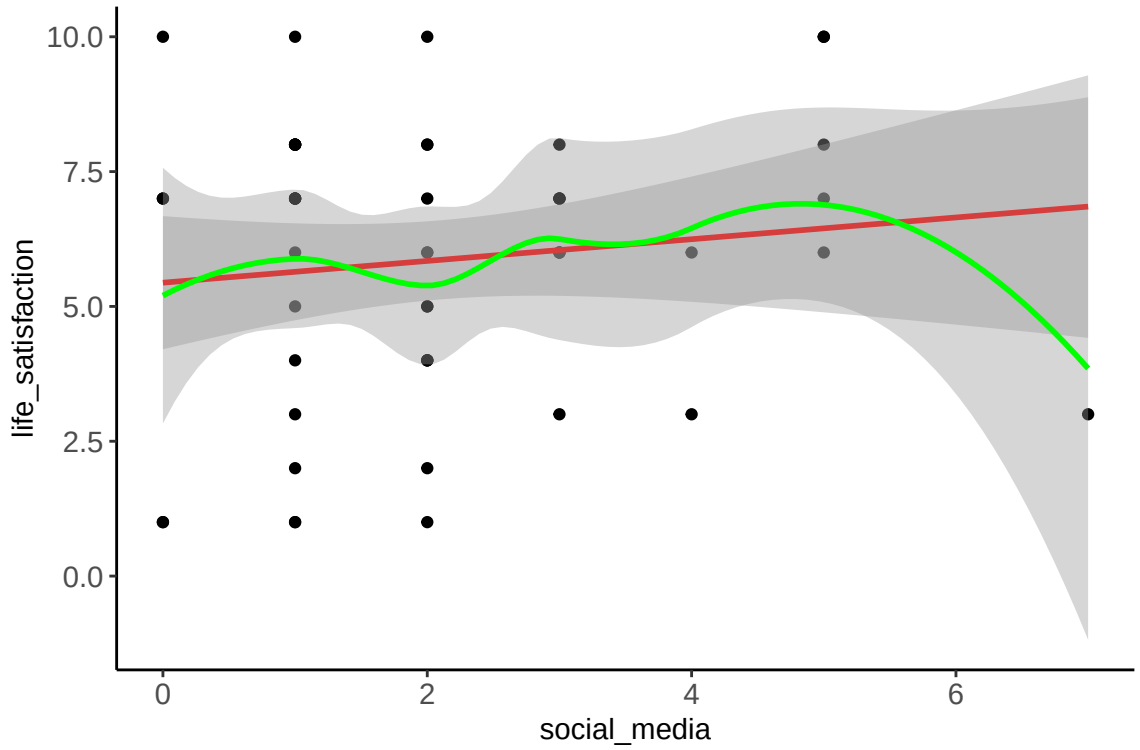

```
data |> plot_scatter("exercise", "life_satisfaction")
```



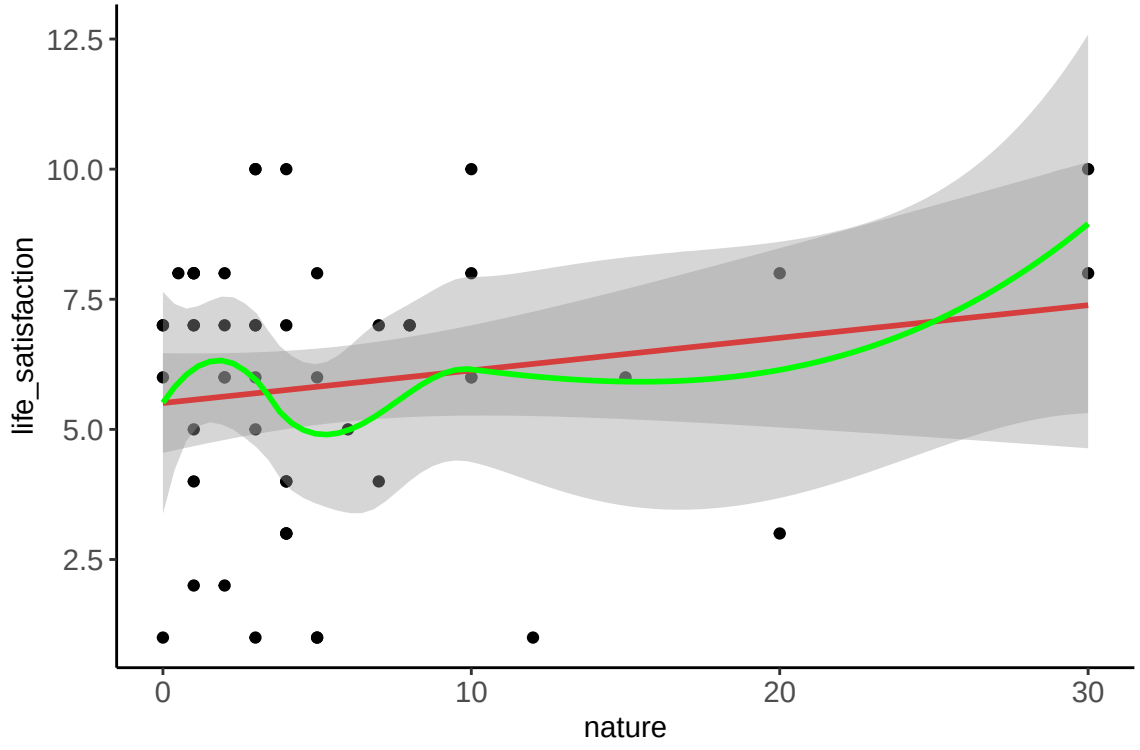

```
data |> plot_scatter("mean_conv", "life_satisfaction")
```



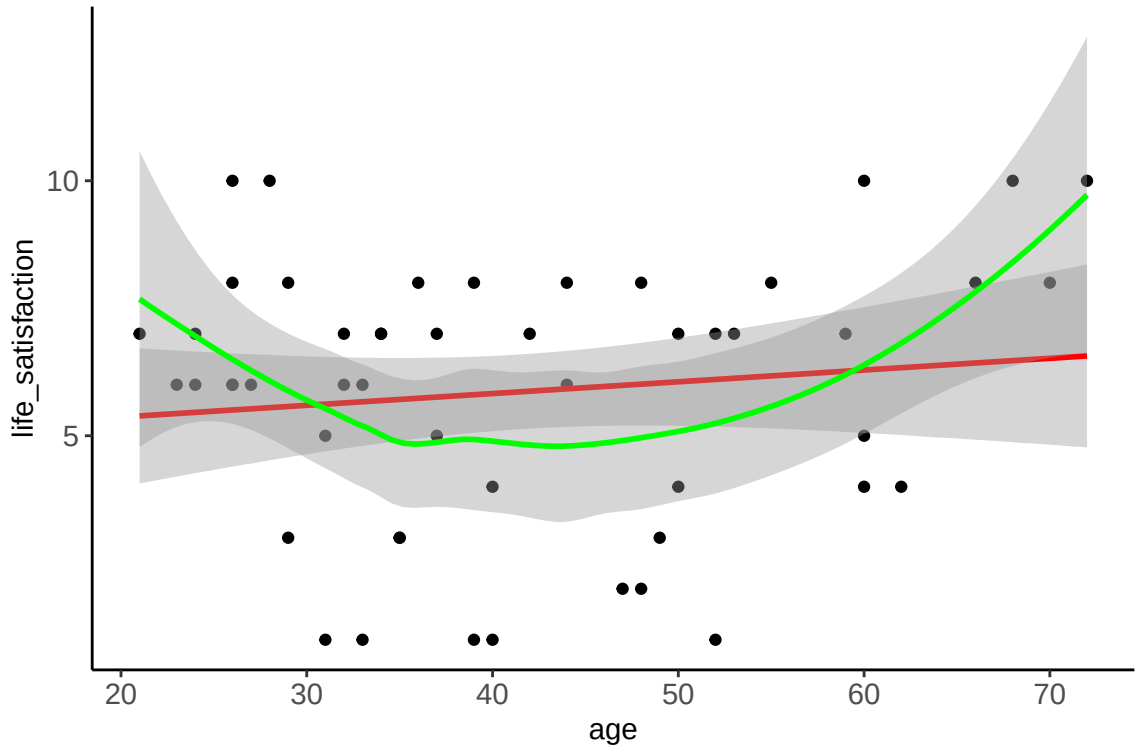

```
data |> plot_scatter("social_media", "life_satisfaction")
```




```
data |> plot_scatter("nature", "life_satisfaction")
```

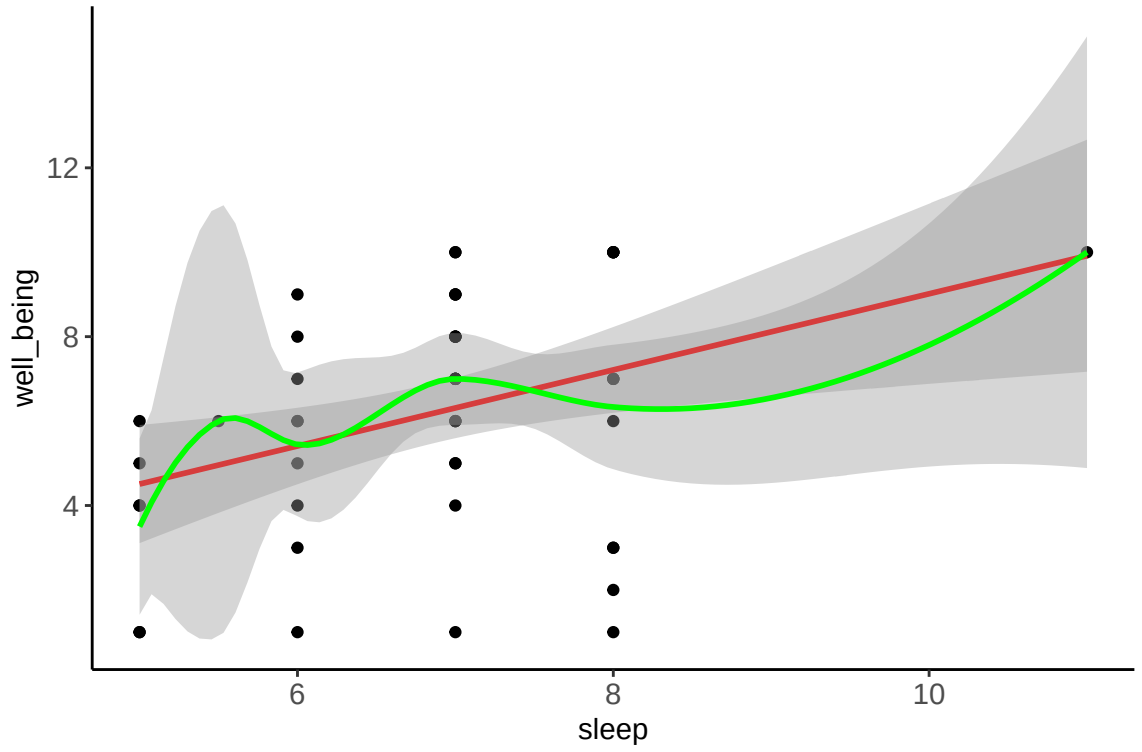



```
data |> plot_scatter("age", "life_satisfaction")
```

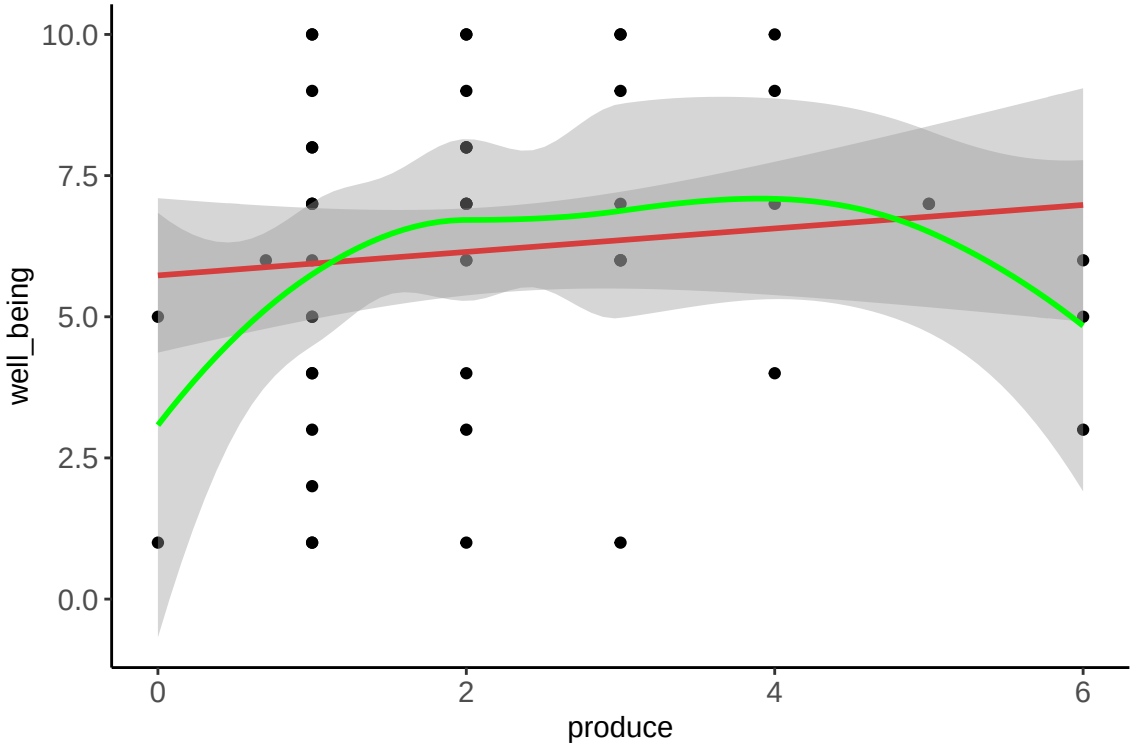


Here, scatter plots were created comparing well-being with numeric independent variables. These plots essentially reinforce the observations made above.

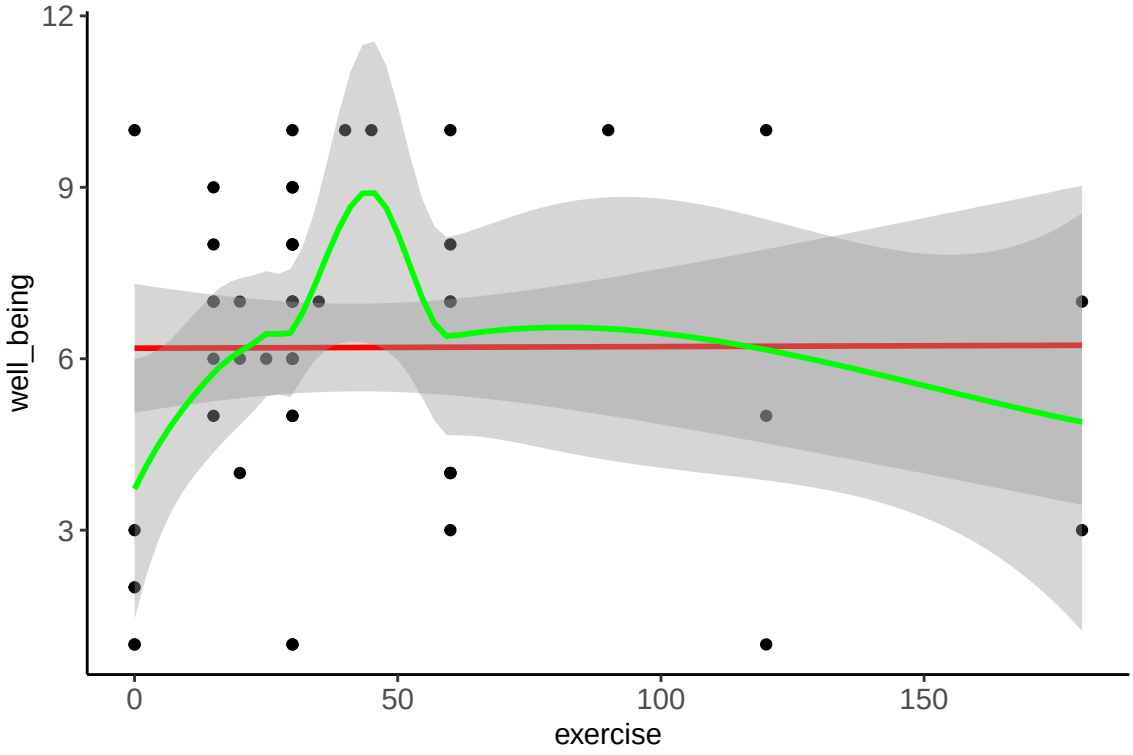
```
# scatter plots of independent variables against 'well_being'
data |> plot_scatter("sleep", "well_being")
```



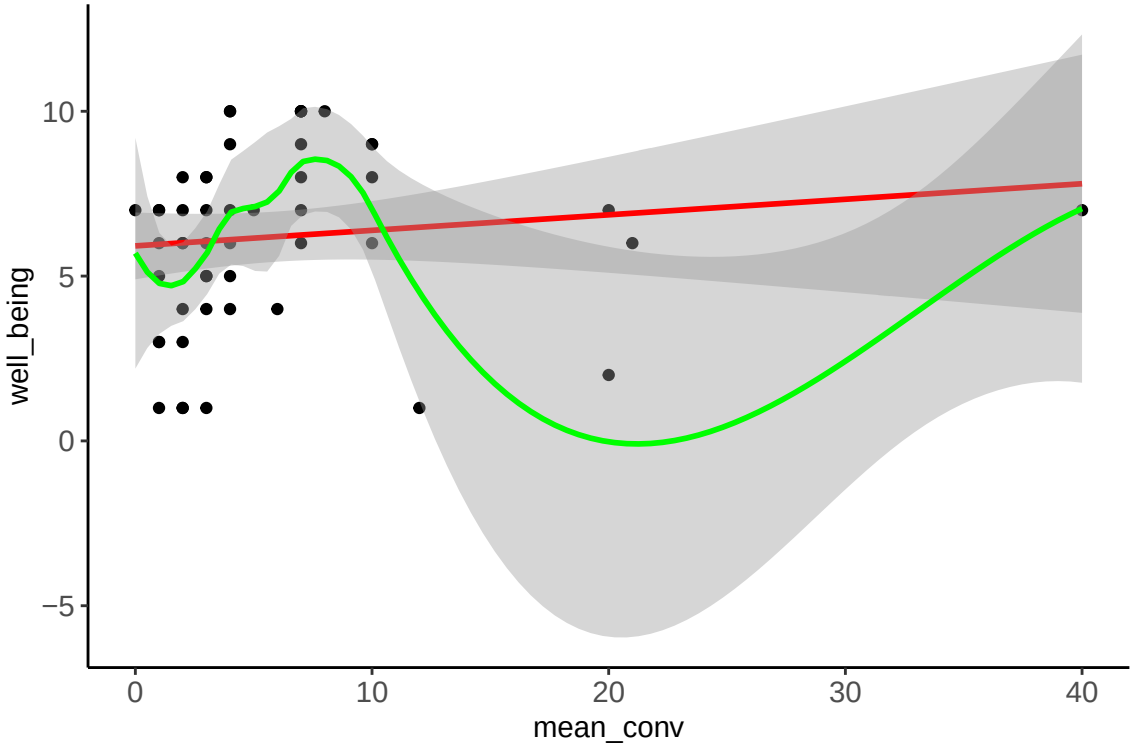
```
data |> plot_scatter("produce", "well_being")
```



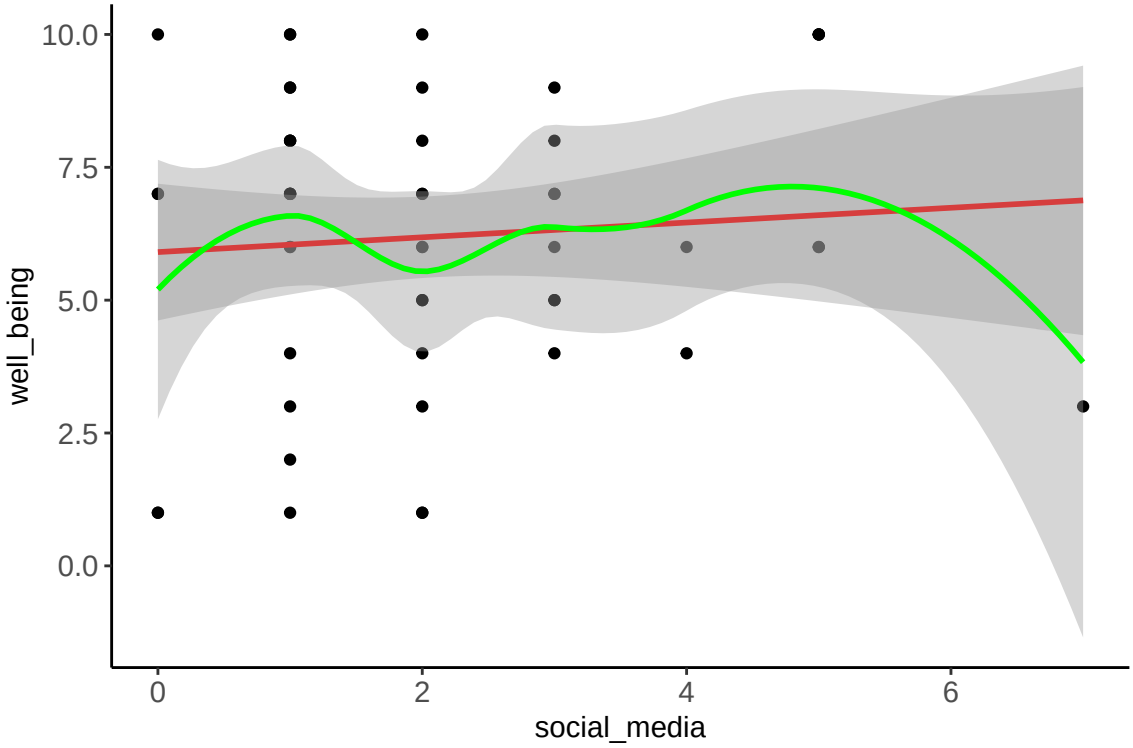

```
data |> plot_scatter("exercise", "well_being")
```



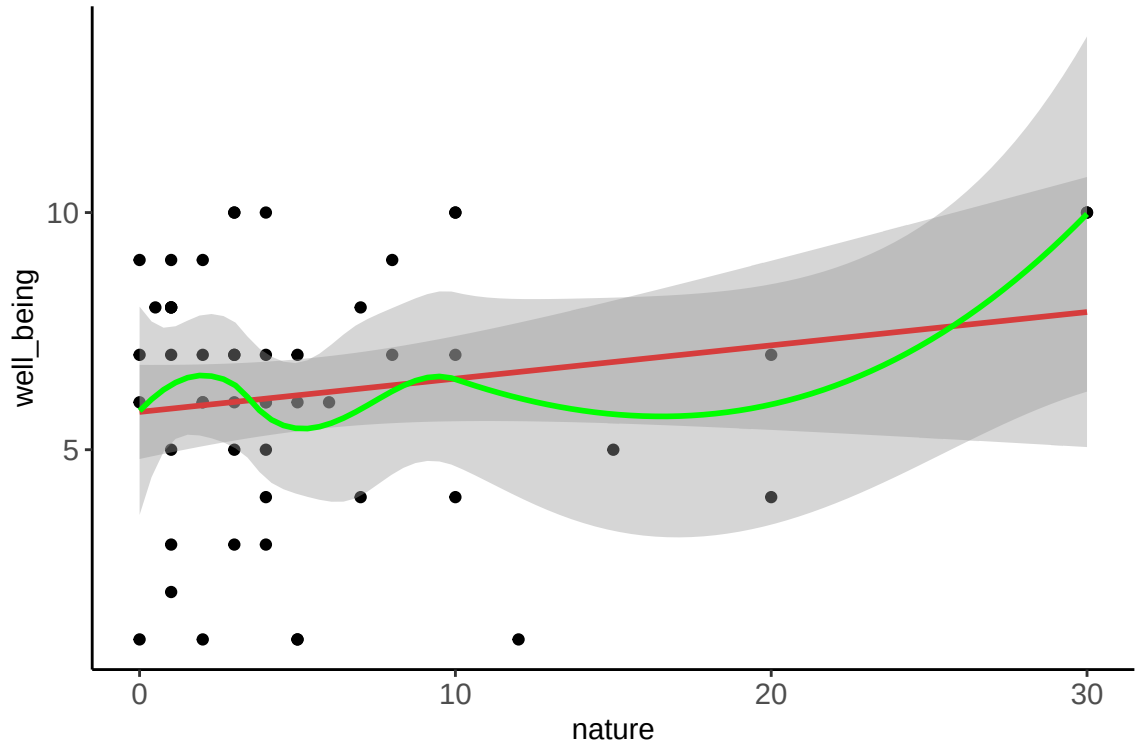
```
data |> plot_scatter("mean_conv", "well_being")
```



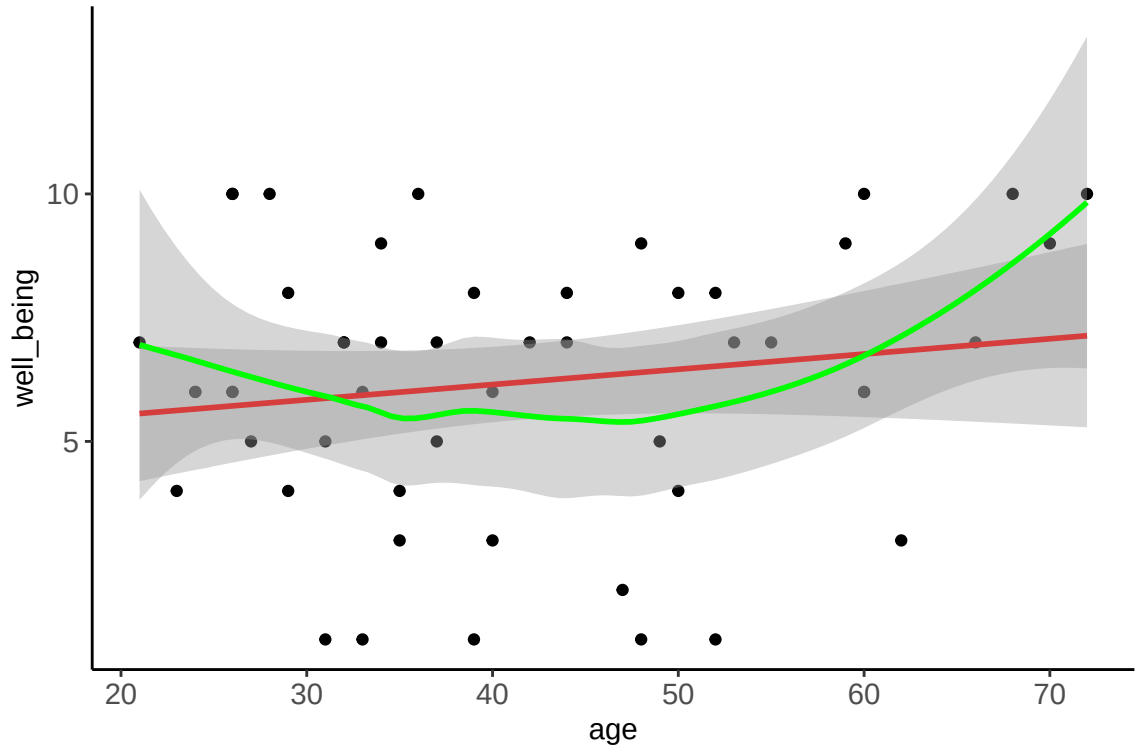

```
data |> plot_scatter("social_media", "well_being")
```



```
data |> plot_scatter("nature", "well_being")
```

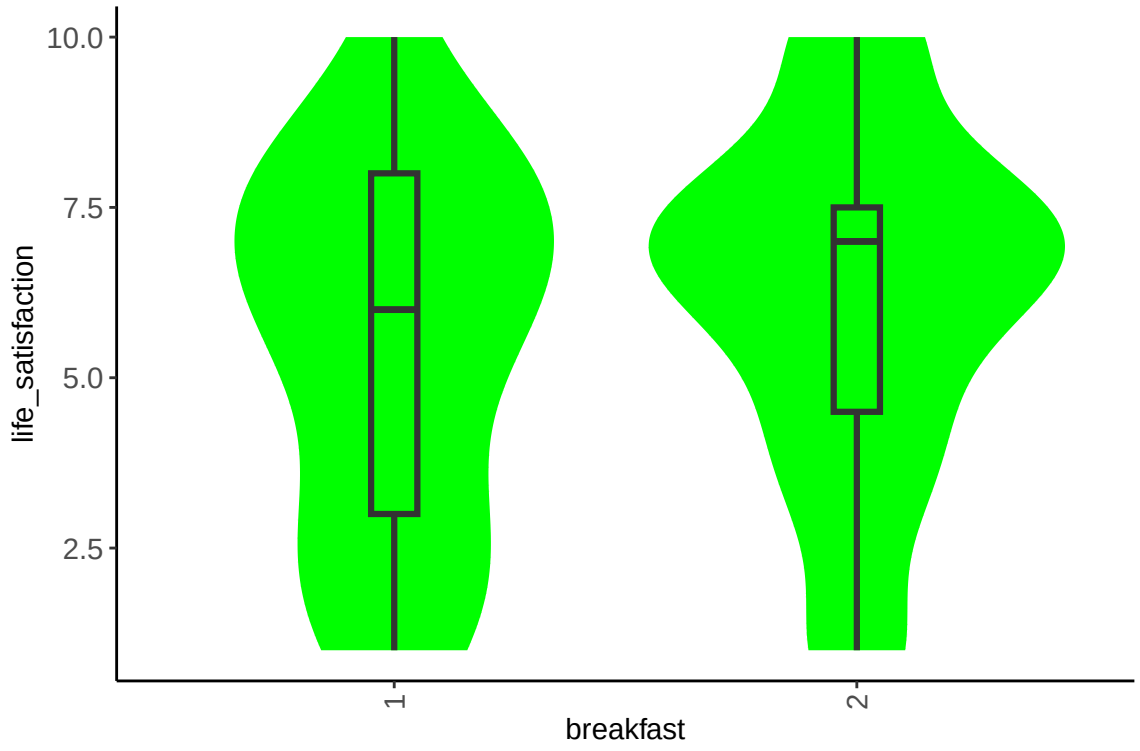



```
data |> plot_scatter("age", "well_being")
```

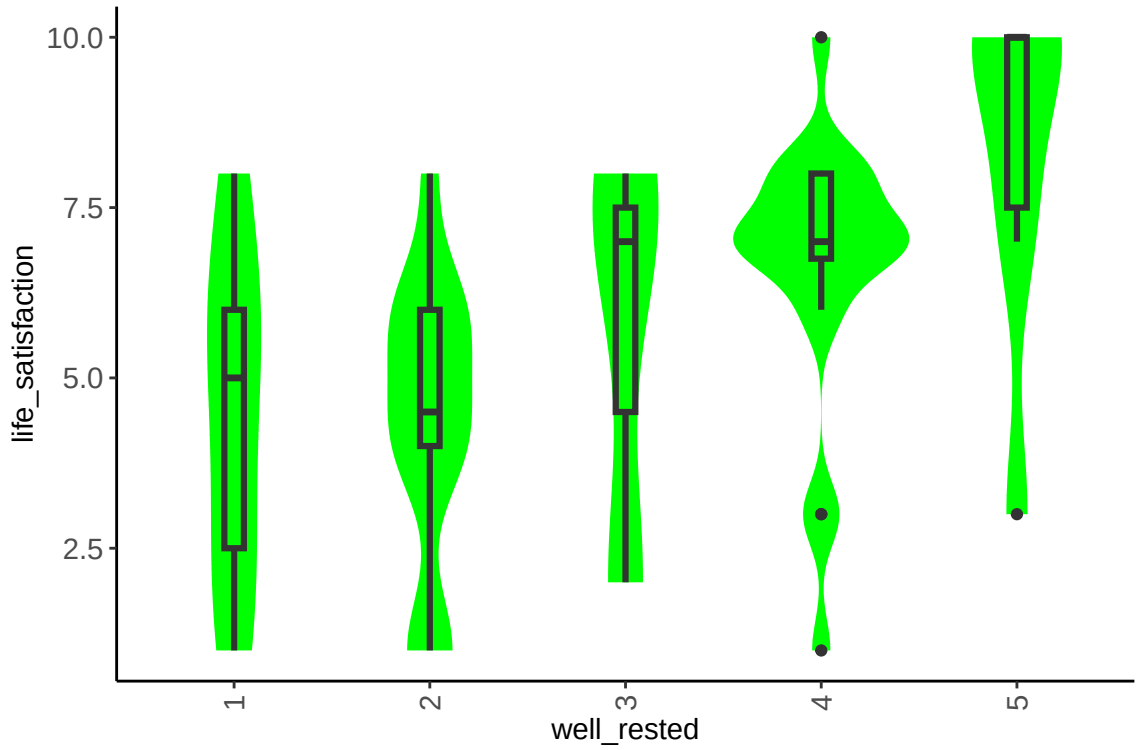


Next, grouped violin plots were created comparing life satisfaction with categorical independent variables. There appears to be a very slight difference in life satisfaction between breakfast-eaters and non-breakfast-easters, with those who consume breakfast reported slightly greater life satisfaction. Feeling a sense of belonging may have a correlation with life satisfaction, as those who agreed more strongly with that statement reported greater life satisfaction. Those who reported feeling under pressure more often typically reported lower life satisfaction, and those who can manage their stress more effectively typically reported greater life satisfaction.

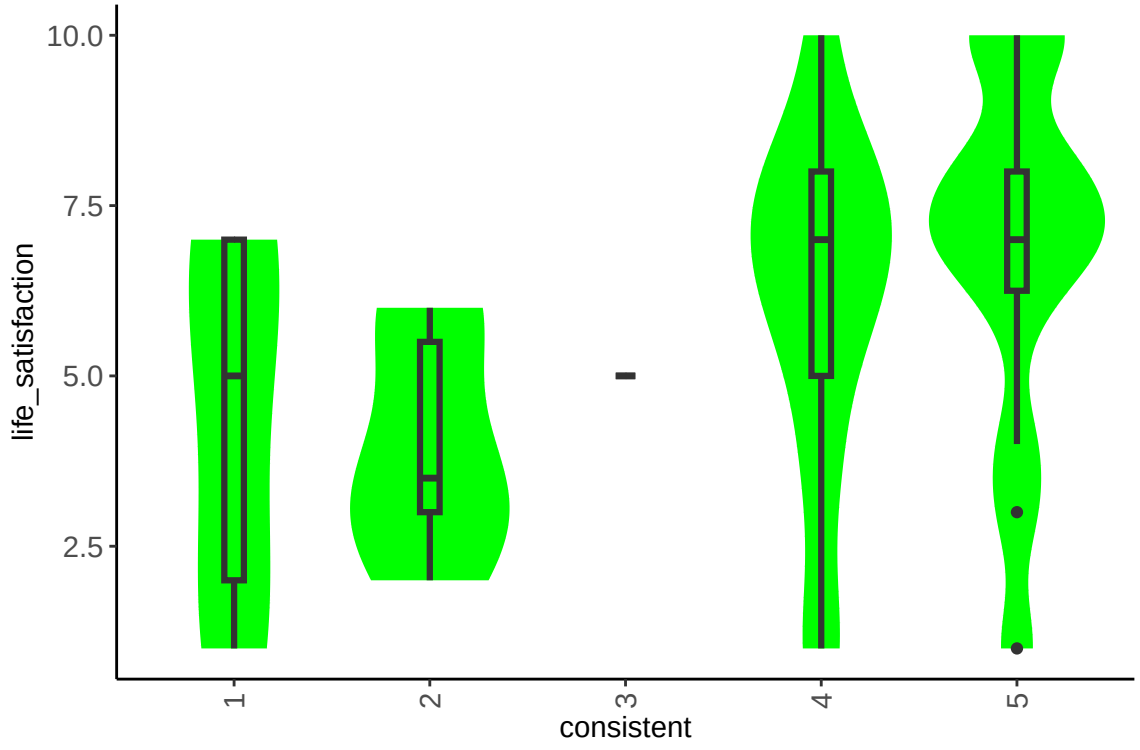
```
# grouped violin plots of factor variables against 'life_satisfaction'
data |> plot_grouped_box_violin("breakfast", "life_satisfaction")
```



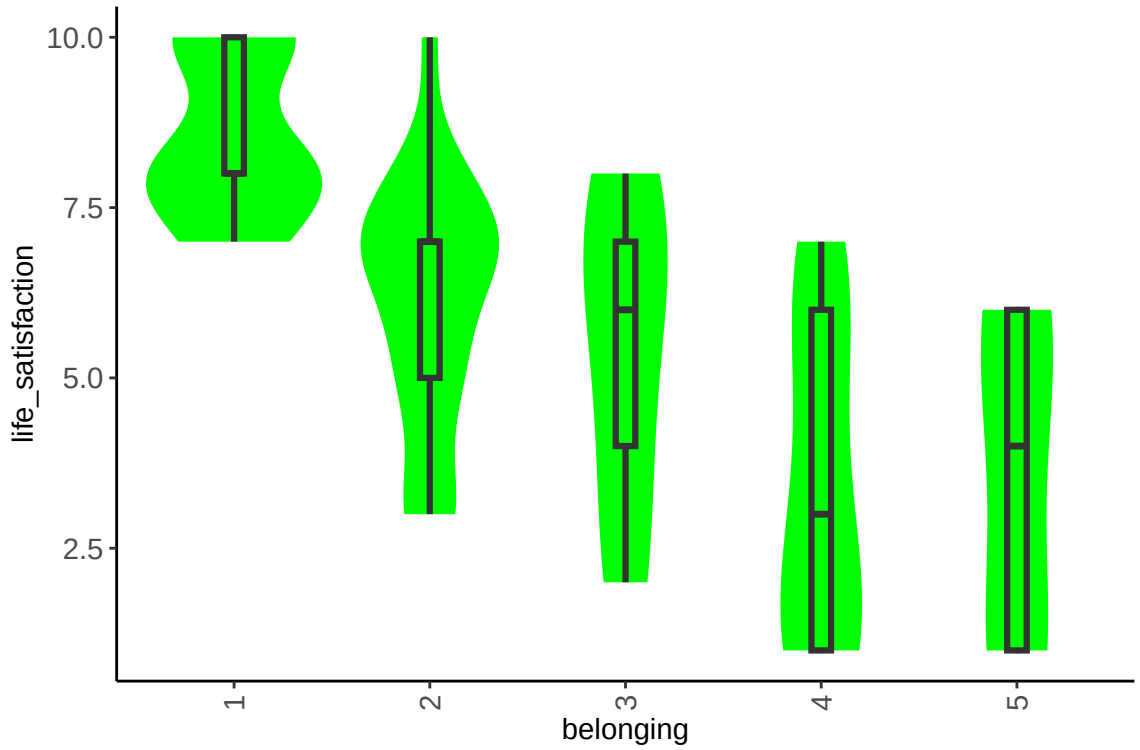

```
data |> plot_grouped_box_violin("well_rested", "life_satisfaction")
```



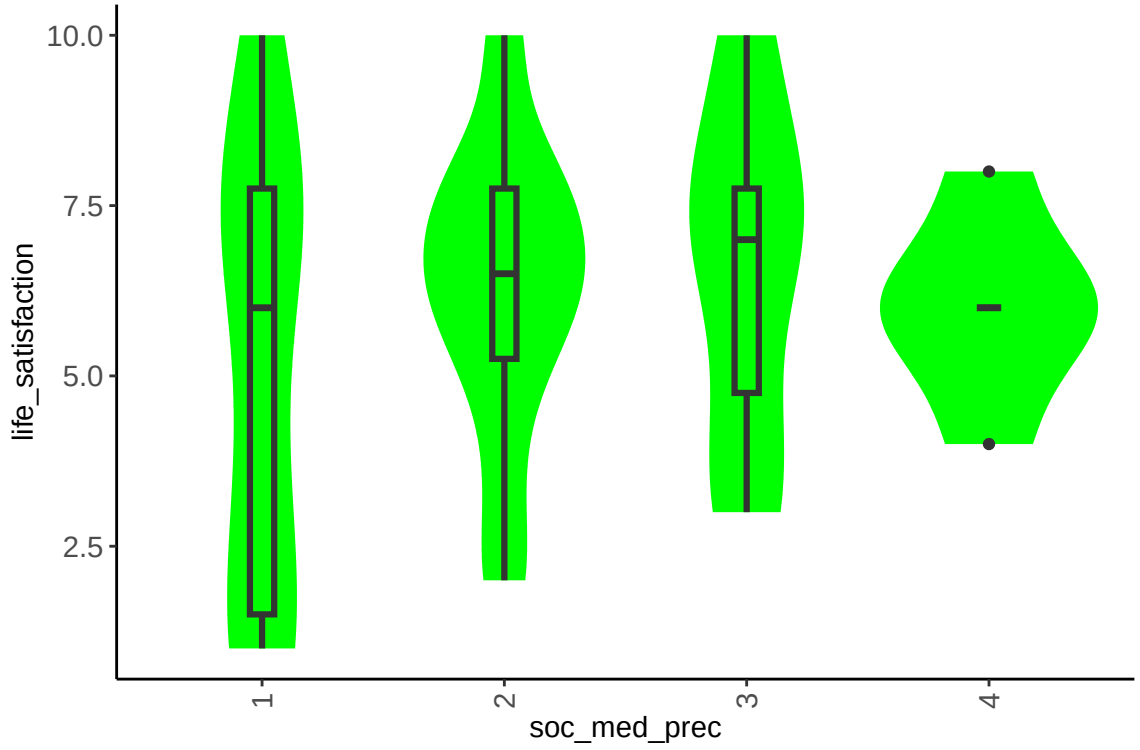

```
data |> plot_grouped_box_violin("consistent", "life_satisfaction")
```



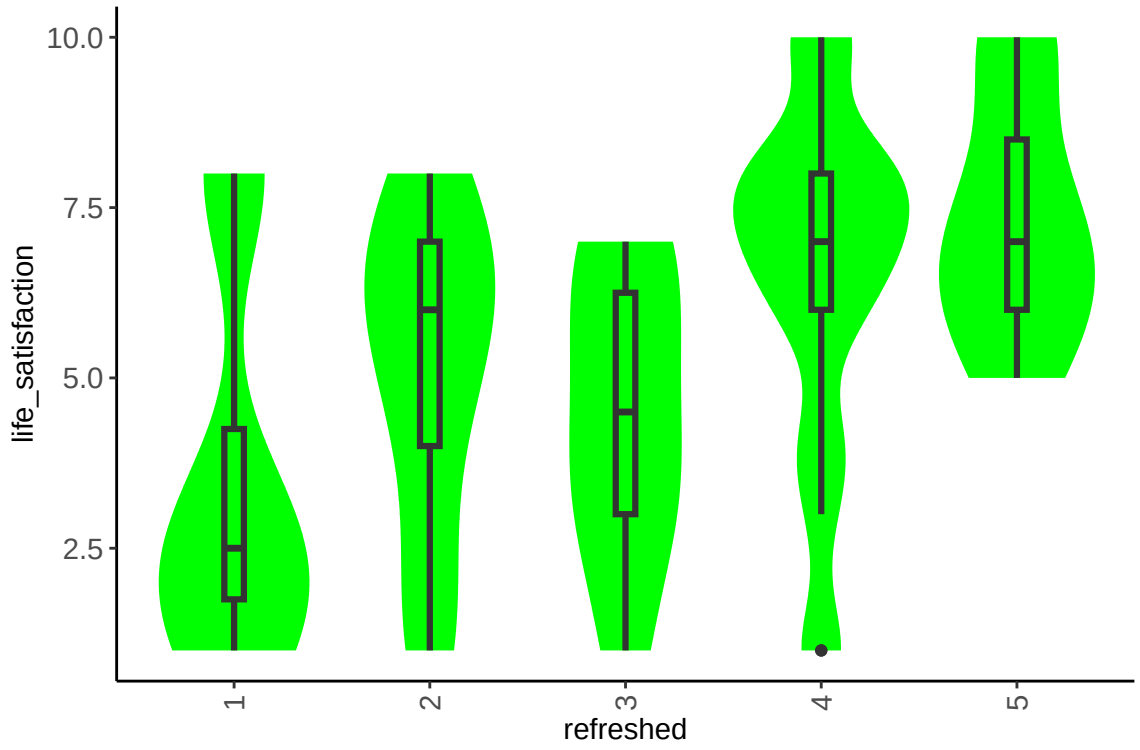

```
data |> plot_grouped_box_violin("belonging", "life_satisfaction")
```



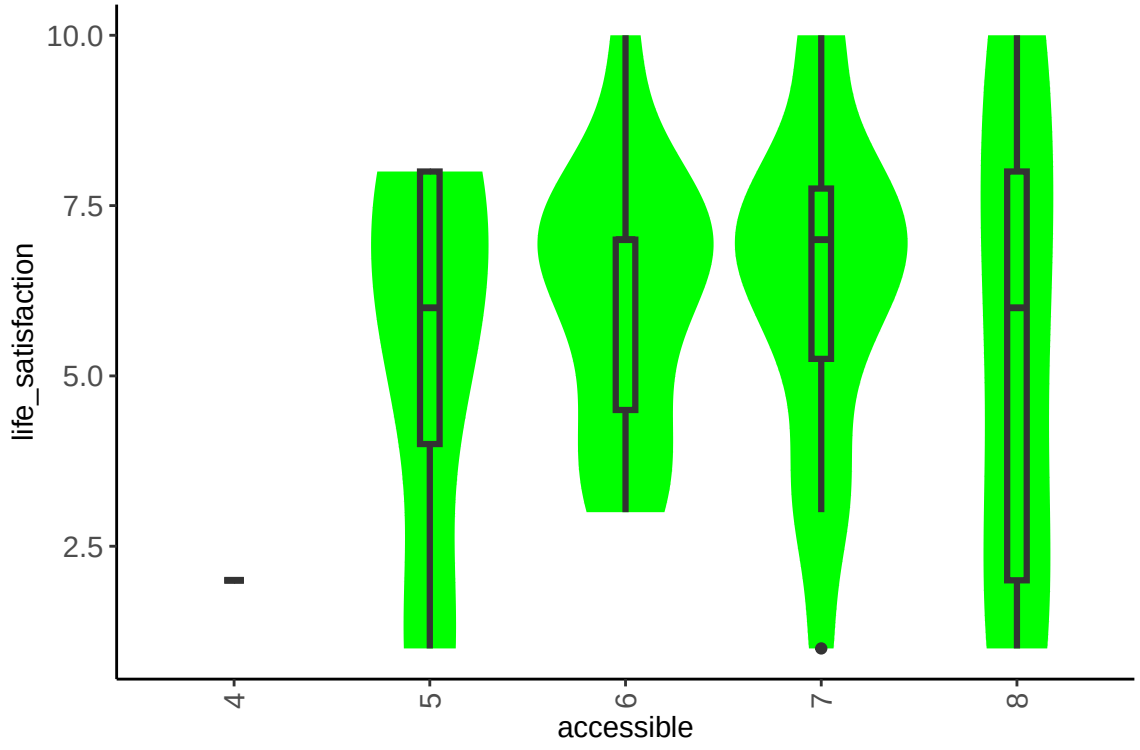

```
data |> plot_grouped_box_violin("soc_med_prec", "life_satisfaction")
```



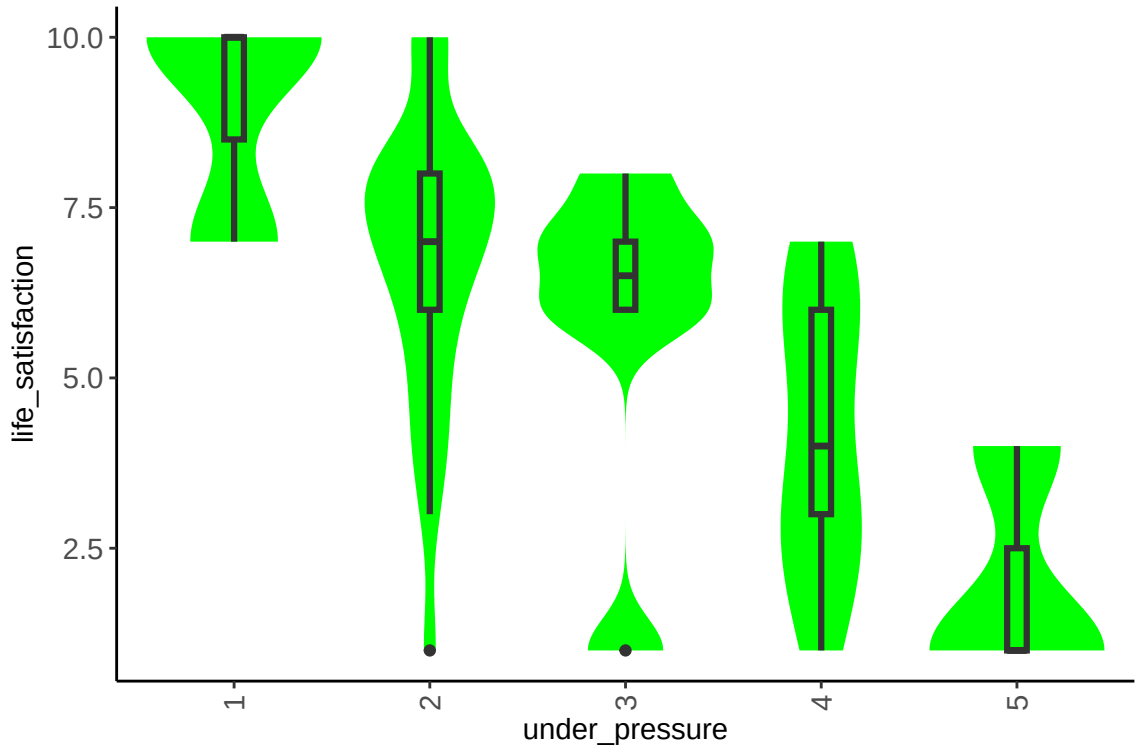

```
data |> plot_grouped_box_violin("refreshed", "life_satisfaction")
```



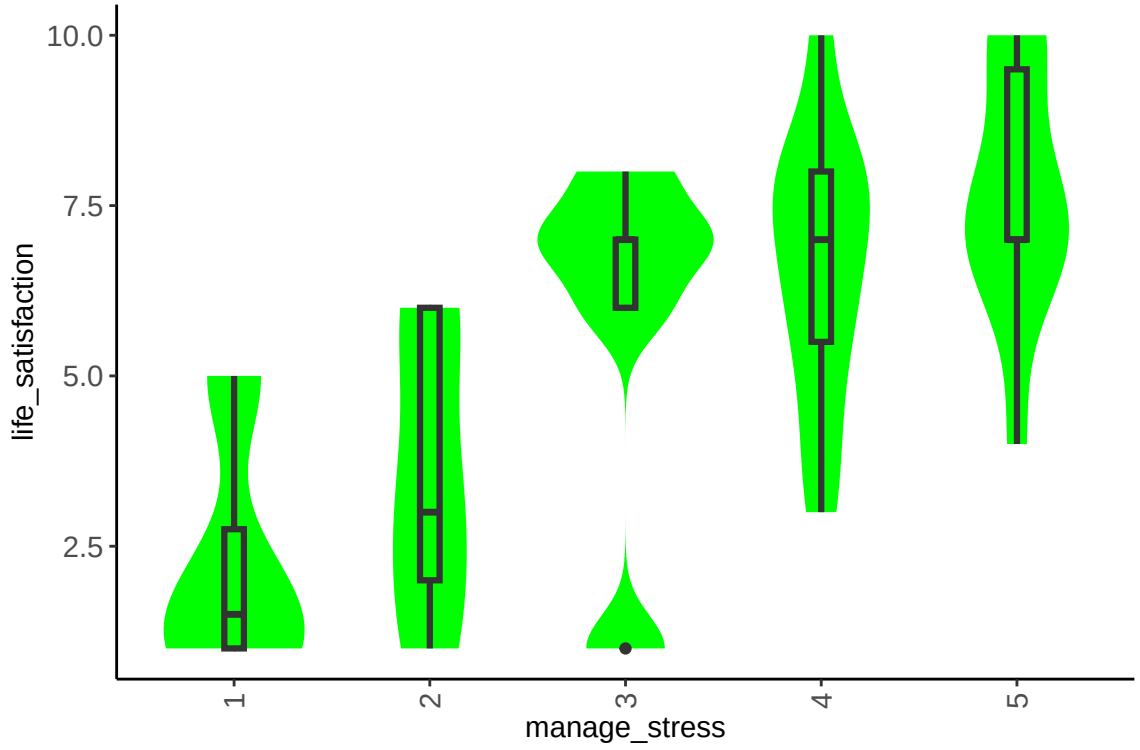

```
data |> plot_grouped_box_violin("accessible", "life_satisfaction")
```




```
data |> plot_grouped_box_violin("under_pressure", "life_satisfaction")
```

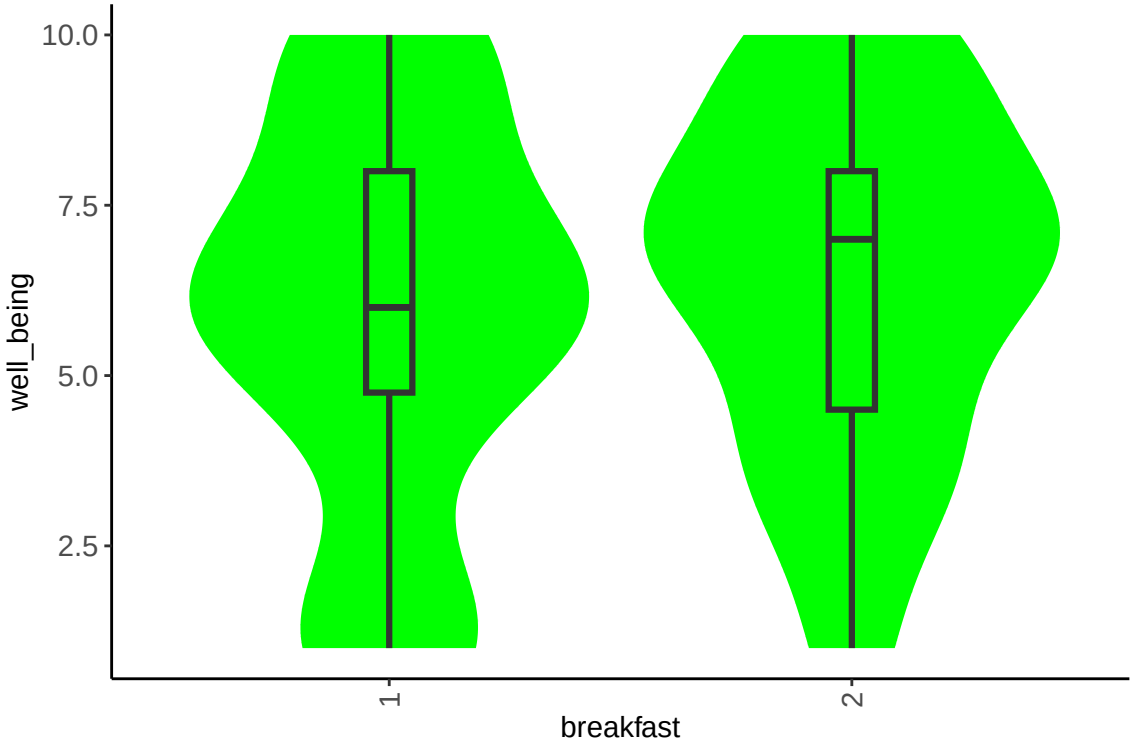



```
data |> plot_grouped_box_violin("manage_stress", "life_satisfaction")
```

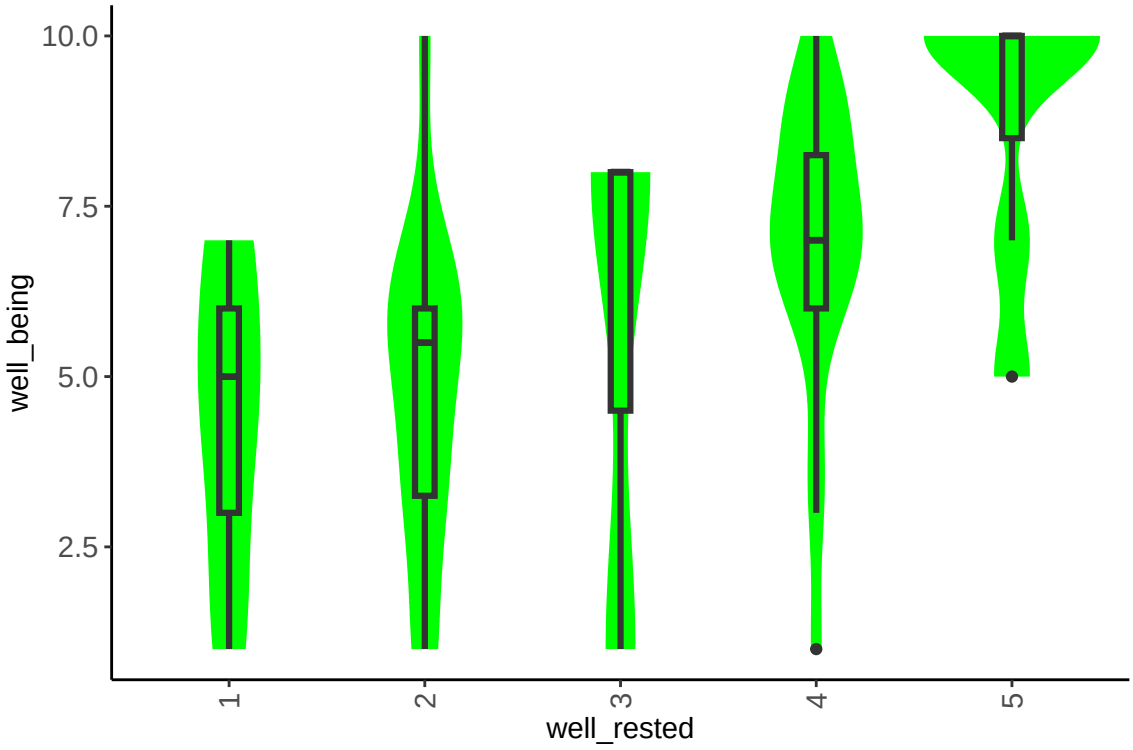


In grouped violin plots comparing well-being with categorical independent variables, the same observations from above were noted.

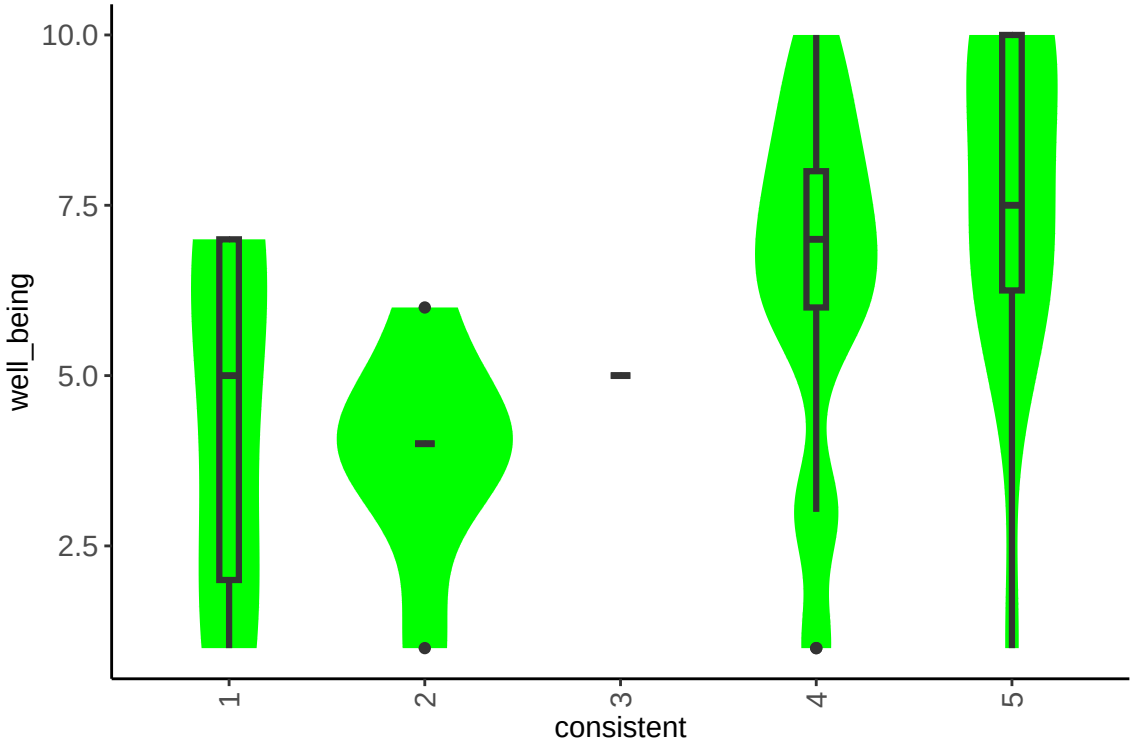
```
# grouped violin plots of factor variables against 'well_being'
data |> plot_grouped_box_violin("breakfast", "well_being")
```



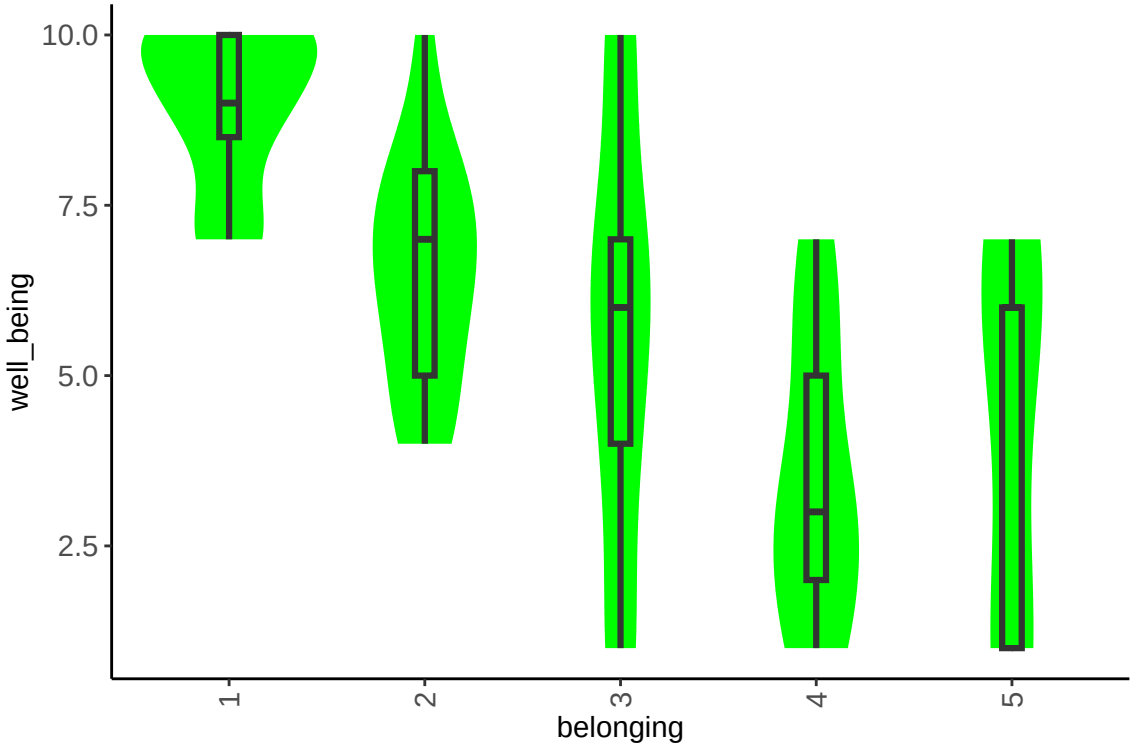

```
data |> plot_grouped_box_violin("well_rested", "well_being")
```



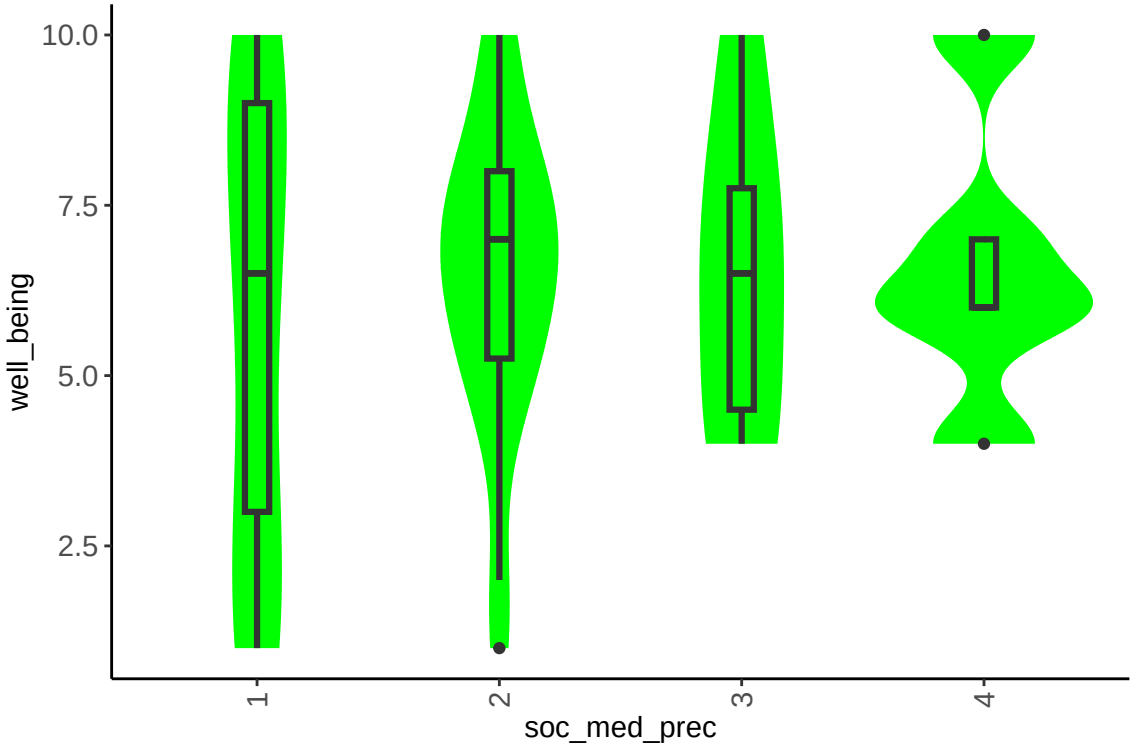
```
data |> plot_grouped_box_violin("consistent", "well_being")
```



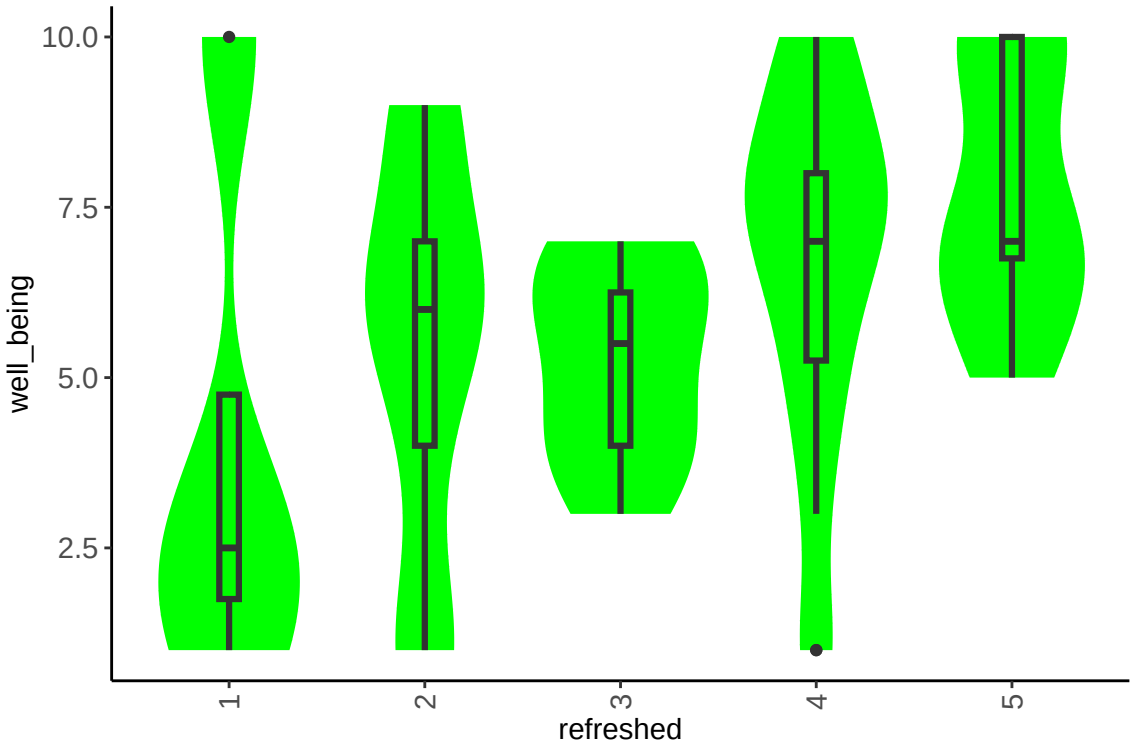

```
data |> plot_grouped_box_violin("belonging", "well_being")
```



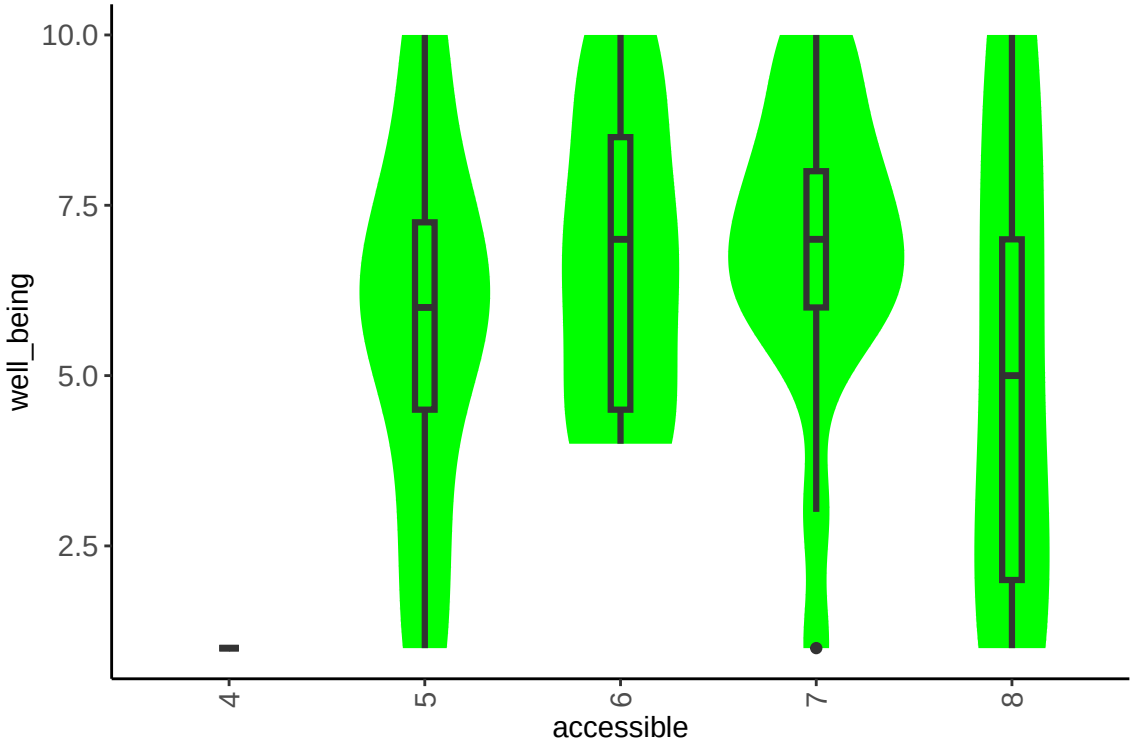
```
data |> plot_grouped_box_violin("soc_med_prec", "well_being")
```



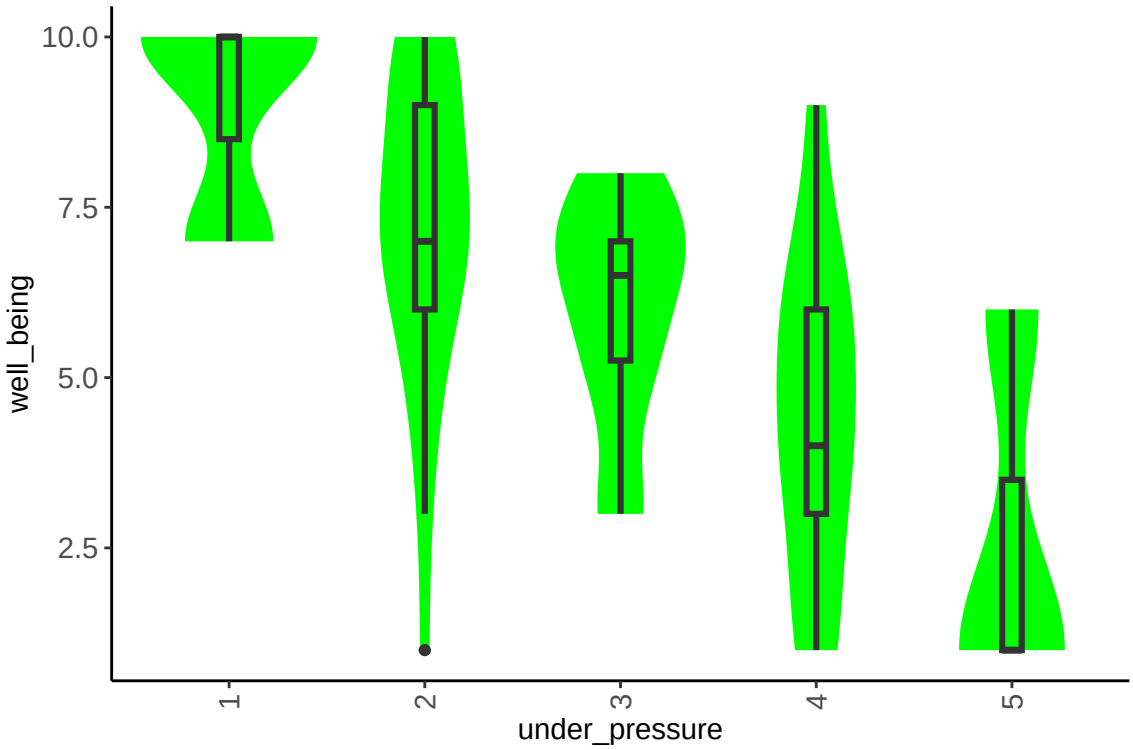

```
data |> plot_grouped_box_violin("refreshed", "well_being")
```



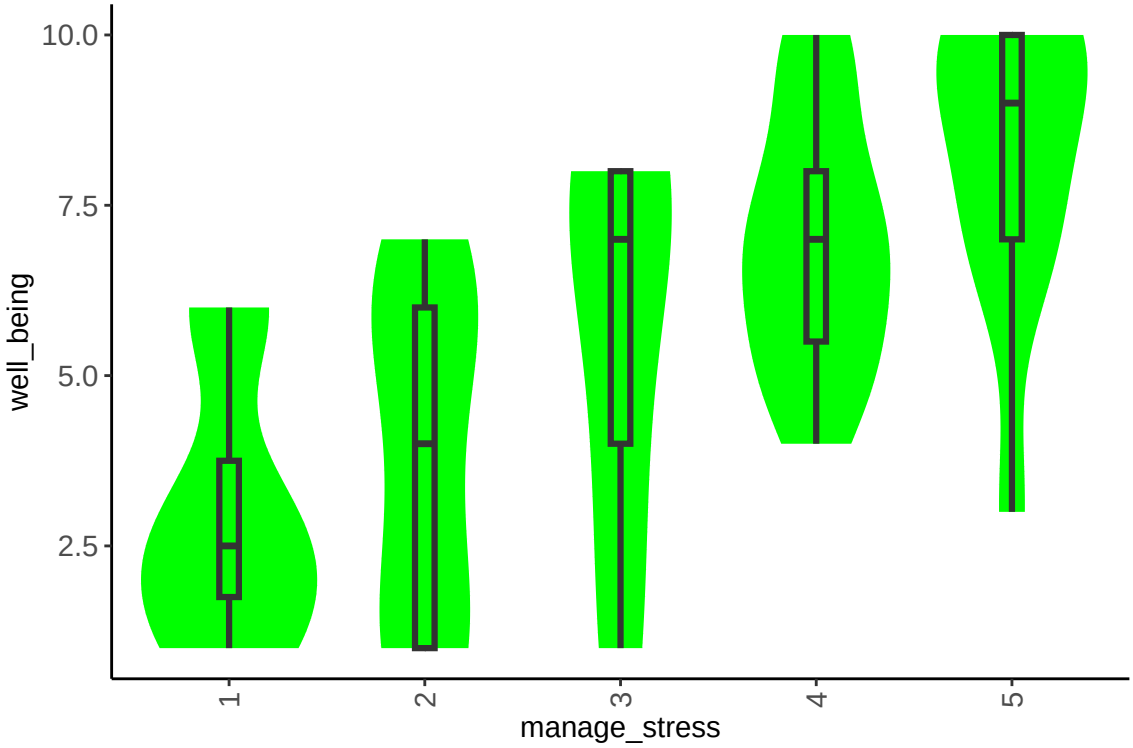
```
data |> plot_grouped_box_violin("accessible", "well_being")
```




```
data |> plot_grouped_box_violin("under_pressure", "well_being")
```



```
data |> plot_grouped_box_violin("manage_stress", "well_being")
```



3 Results

3.1 Linear Model

Many linear models were fit in the process of fitting a linear regression for this study. First, a mean-only model (m_00) of life satisfaction was created. This null model serves as a baseline point assessing average well-being across all participants. The model revealed that the mean well-being score was 6.20, which was significantly greater than zero, $t(50) = 16.40$, $p < .001$.

```
# mean-only model
m_00 <- lm(well_being ~ 1, data = data)
summary(m_00)
```

```
Call:
lm(formula = well_being ~ 1, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-5.1961 -1.6961  0.8039  1.8039  3.8039

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.1961     0.3778    16.4   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 2.698 on 50 degrees of freedom
```

Next, a multiple regression model (m_01) tested the effects of all survey items on participants' well-being. The model was statistically significant, $F(38, 12) = 7.60$, $p < .001$, and explained 96% of the variance in well-being ($R^2 = .9601$, Adjusted $R^2 = .8338$). Feeling well-rested and (somehow) giving social media precedence over necessary tasks were significantly associated with higher well-being, while poorer stress management predicted lower well-being. There were also marginal effects suggesting that lower sense of belonging and reduced access to nature might be related to reduced well-being. However, this model is likely overfit and has multicolinearity within its regressors.

```
# all-inclusive model
m_01 <- lm(well_being ~ sleep + well_rested + produce + processed_food + breakfast + exercise + belongi
summary(m_01)
```

```
Call:
lm(formula = well_being ~ sleep + well_rested + produce + processed_food +
    breakfast + exercise + belonging + mean_conv + social_media +
    soc_med_prec + nature + refreshed + accessible + under_pressure +
    manage_stress, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.49445 -0.27998  0.06619  0.33301  0.94900

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.7435431  5.3716425   0.138   0.8922
sleep            0.0804840  0.3937971   0.204   0.8415
well_rested2     1.5154061  0.8626859   1.757   0.1044
well_rested3     4.8561117  1.6121808   3.012   0.0108 *
well_rested4     2.2553963  0.9757194   2.312   0.0394 *
well_rested5     2.1434686  1.3743018   1.560   0.1448
produce          0.0779446  0.1792348   0.435   0.6714
processed_food2 -0.2955964  1.1590592  -0.255   0.8030
processed_food3 -1.2769936  1.4024343  -0.911   0.3805
processed_food4  1.6497114  1.5414033   1.070   0.3056
processed_food5  1.2166022  1.6136347   0.754   0.4654
breakfast2      -0.3678939  0.5685967  -0.647   0.5298
exercise        -0.0087571  0.0071788  -1.220   0.2460
belonging2      -0.7027770  0.7629986  -0.921   0.3752
belonging3      -2.2255979  1.2080086  -1.842   0.0903 .
belonging4      -2.2333352  1.0656235  -2.096   0.0580 .
belonging5      -2.3666702  1.2533286  -1.888   0.0834 .
mean_conv        0.0918955  0.0461382   1.992   0.0697 .
social_media     0.0009743  0.1909614   0.005   0.9960
soc_med_prec2    0.9234349  0.7252466   1.273   0.2270
soc_med_prec3    0.3685977  0.7776703   0.474   0.6440
soc_med_prec4    3.9803303  1.1419594   3.486   0.0045 **
nature           0.0668311  0.0420601   1.589   0.1381
refreshed2       0.8623764  1.2853566   0.671   0.5150
refreshed3      -1.2746218  1.3437922  -0.949   0.3616
refreshed4       1.6171832  1.4295358   1.131   0.2800
refreshed5       2.0603895  1.4737218   1.398   0.1874
accessible5      2.6614503  2.7354530   0.973   0.3498
```

```
accessible6      4.4471510  2.6709420   1.665   0.1218
accessible7      4.8428232  2.5829944   1.875   0.0853 .
accessible8      3.2282648  2.5729568   1.255   0.2335
under_pressure2 -1.4532343  1.3635684  -1.066   0.3075
under_pressure3  1.2005861  1.8585264   0.646   0.5304
under_pressure4 -2.3125927  1.7379275  -1.331   0.2080
under_pressure5 -1.7841334  2.2448406  -0.795   0.4422
manage_stress2  -1.7033957  1.3534408  -1.259   0.2321
manage_stress3  -3.5001899  1.5676543  -2.233   0.0454 *
manage_stress4  -0.3918739  1.3491769  -0.290   0.7764
manage_stress5   1.0541206  1.3298756   0.793   0.4434
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.1 on 12 degrees of freedom
Multiple R-squared:  0.9601,    Adjusted R-squared:  0.8338
F-statistic: 7.602 on 38 and 12 DF,  p-value: 0.0002908
```

Based on results from the above model, a third model (m_02) was created including key predictors of well-being in this dataset. The model was statistically significant, $F(9, 41) = 10.7$, $p < .001$, indicating that the predictors collectively explained a substantial proportion of variance in life satisfaction. The model accounted for approximately 70.1% of the variance in life satisfaction ($R^2 = .701$, Adjusted $R^2 = .636$). While this model explains less variance than the all-inclusive model (96% vs. 70%), it takes care to factor in key predictors of well-being only. Overall, this model has greater significance, less overfitting, and no multicolinearity between its regressors.

```r
# correlated features model
m_02 <- lm(well_being ~ sleep + belonging + manage_stress, data = data)
summary(m_02)
```

```
Call:
lm(formula = well_being ~ sleep + belonging + manage_stress,
    data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-3.8239 -0.7589 -0.0002  0.7797  3.2247

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)     0.2328     1.8784   0.124 0.901979
sleep           0.8891     0.2246   3.958 0.000293 ***
belonging2     -1.3702     0.6539  -2.095 0.042347 *
belonging3     -2.7043     0.9037  -2.992 0.004670 **
belonging4     -4.0738     0.8528  -4.777 2.29e-05 ***
belonging5     -2.5312     0.9853  -2.569 0.013939 *
manage_stress2  0.1823     0.9965   0.183 0.855740
manage_stress3  1.3679     1.1182   1.223 0.228189
manage_stress4  2.1337     0.9631   2.216 0.032332 *
manage_stress5  2.6542     1.0696   2.481 0.017275 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.628 on 41 degrees of freedom
```

```
Multiple R-squared:  0.7014,    Adjusted R-squared:  0.6358
F-statistic:  10.7 on 9 and 41 DF,  p-value: 2.594e-08
```

4 Discussion

The present study examined how sleep quality, sense of belonging, and stress management skills predict life satisfaction. The regression model was statistically significant and explained a large proportion of the variance in life satisfaction ($R^2 = .70$), suggesting that these three factors together are robust predictors of overall life satisfaction.

Consistent with prior research, better sleep was strongly and positively associated with greater life satisfaction. This finding highlights the critical role of sleep in promoting well-being and aligns with previous literature linking sleep quality to improved mood and well-being (Weinberg et al., 2016).

Sense of belonging was also a significant predictor of well-being. Participants who reported lower levels of belonging consistently reported lower well-being. This is fairly in line with pervious research, which suggests that feelings of loneliness may have negative impacts on well-being (Wang et al., 2023). Interestingly, the magnitude of the negative relationship increased as levels of belonging decreased, suggesting a dose-response effect: the less one feels a sense of belonging, the lower their reported well-being.

Finally, the ability to manage stress emerged as a significant positive predictor of well-being at higher levels. Participants who endorsed stronger stress management skills reported significantly greater life satisfaction. This reinforces the importance of emotional regulation and coping mechanisms in enhancing well-being, particularly in the face of daily stressors and challenges.

These findings have practical implications for mental health interventions and wellness programming. Interventions aimed at improving sleep hygiene, strengthening social connectedness, and teaching effective stress management techniques may be particularly beneficial in enhancing life satisfactionvand well-being. For instance, programs that combine cognitive-behavioral strategies for sleep with group-based activities that foster belonging may have even more enhanced effects.

Despite the strengths of the study, several limitations should be noted. The cross-sectional nature of the data precludes any causal inferences; we cannot determine whether better sleep, belonging, or stress management cause improved well-being or vice versa. In addition, self-report measures may be subject to social desirability or recall bias. Future research could benefit from longitudinal designs and more objective assessments of sleep and stress regulation.

This study provides strong evidence that sleep, social belonging, and stress management are key contributors to life satisfaction. Efforts to improve individual and community well-being should prioritize these modifiable lifestyle and psychosocial factors.

References

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Muscaritoli, M. (2021). The impact of nutrients on mental health and well-being: Insights from the literature. *Frontiers in Nutrition*, *8*.

Wang, X., Wu, Y., Shi, X., Chen, Y., Xu, Y., Xu, H., Ma, Y., & Zang, S. (2023). Associations of lifestyle with mental health and well-being in chinese adults: A nationwide study. *Frontiers in Nutrition*, *10*.

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